國　立　中　央　大　學

光電科學與工程學系

博　士　論　文

Integrated Periodically and Aperiodically Poled Lithium Niobate (PPLN/APLN) Photonics and Laser Devices

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Abstract

Compact, multi-function yet efficient optical devices are in great demand in many photonics and laser systems and applications. The ingenious integration of the unique electro-optic (EO) effects of a quasi-phase-matching (QPM) material with its capability of performing efficient nonlinear frequency conversion has led to the development of many attractive monolithic multi-function devices. To continue and extend these efforts and achievements, several advanced integrated photonics and laser devices of capable of performing multiple optical functions based on periodically and aperiodically poled lithium niobate (PPLN/APLN) devices were studied and demonstrated in this dissertation work. They were realized upon the idea of manipulation of the engineerable domain structure of a QPM crystal to optimize performing the prescribed EO and nonlinear-optic (NLO) processes simultaneously to achieve the multi-function operation in photonics and laser systems. These devices are of potential for applications in optical communications, optical storages, biomedicine, displays, remote sensing, etc.

In chapter 1, the motivation and background introduction of this study will be given, while in chapter 2, the theory and working principles of EO QPM devices will be introduced. In chapter 3, the design, construction, and experimental demonstration of a laser-diode-pumped, electro-optically internal-Q-switched laser system radiating at 1.085 μm fabricated using a periodically poled Nd:MgO:$LiNbO_3$ (Nd:MgO:PPLN) crystal will be detailed. In chapter 4, a design scheme for and the first experimental demonstration of active narrowband multi-wavelength filters based on APLN crystal will be presented. In chapter 5, a novel electro-optically Q-switched intracavity second-harmonic generation (SHG) Nd:YVO_4 laser by using a single-grating-structure APLN crystal optimized for simultaneously performing a laser Q-switch and a second-harmonic generator will be revealed and characterized. In chapter 6, a gain-enhanced and spectral-narrowed optical parametric oscillator based on a monolithic PPLN integrating an optical parametric generator with two electro-

optically active polarization-mode converters will be reported. Unique spectral manipulation of the OPO signal will also be demonstrated. The summary and outlook of this work will finally be given in chapters 7 and 8, respectively.

The success of these demonstrations will certainly be a significant step in advancing the nonlinear integrated optics essential for various applications. The study and development of several derived and advanced nonlinear integrated photonics and laser devices are under way.

中文摘要

在許多光電和雷射系統應用中需要使用到緊緻、多功能與高效率的光學元件。準相位匹配材料中獨特的積體式電光效應加上本身所具備的高效率非線性頻率轉換能力可用來開發許多引人注目的單塊晶多功能元件。延續過往的研究成果，在本論文中研究與論證了幾個在週期式與非週期式極性反轉鈮酸鋰(PPLN/APLN)中實現的多功能積體式光電與雷射元件。它們都利用準相位匹配晶體中可工程化的鐵電結構用以最佳化並同時實現前述之電光與非線性多功能光電與雷射系統。這些元件有極大的潛力應用在光通訊、光儲存、生醫、顯示技術與遙測等用途上。

於第一章，將會簡介研究動機與背景。於第二章，將會說明電光式準相位匹配之理論與工作原理。於第三章，將會詳細論述利用一塊釹鎂摻雜週期式極性反轉鈮酸鋰(Nd:MgO:PPLN)晶體實現一波長為 1.085 μm 之雷射泵激電光式內部 Q 調制雷射系統之設計、建造與實驗論證之方法與流程。於第四章，首次實驗論證利用一 APLN 晶體實現主動式窄帶多波長濾波器之設計大綱將會被提出。於第五章，將會論證與分析一新穎的電光式 Q 調制腔內二倍頻(SHG)釹釩酸釔雷射，它使用了一塊經過優化設計之單一光柵結構 APLN 晶體，可同時實現一個雷射 Q 開關與一個二倍頻產生器。於第六章，將會提出建立於單塊 PPLN 的一個增益被提高與頻譜被窄化的光參數振盪器(OPO)，其整合了一個光參數產生器與兩個電光式偏極化模態轉換器。文中也會論證其獨特的 OPO 信號頻譜處理能力。此研究工作的總結與未來展望最後會在第七章與第八章分別被提出。

這些成功的論證無疑地是發展非線性積體光學多項實質應用的重大進展。據此進展而研發的數個先進非線性積體光電與雷射元件正持續研究中。

Acknowledgement

一連串的意外穿插在比原本預期還要漫長的博士班生涯中，也讓我這幾年的生活比預期中豐富的太多。

博士班生涯一開始還存有幾年就可以畢業的天真想法，因為原本以為資格考已經是最大的挑戰了。沒想到只花了一年就戰勝資格考的我，卻陷入了單打獨鬥的困境中，上無學長下無學弟。迷惘了兩年，晃到了台大與中研院原分所，見識到了所謂天才的厲害、體驗到了所謂的國際頂尖研討會也認識了一群行為作風與鄉下學校大異其趣的研究生。

回到中央，情況稍有好轉，開始有了可以討論研究的學弟們，有了比自己還熱血的指導教授陳彥宏老師，也算是脫離了單打獨鬥的困境，真正才開始體驗到團隊合作的研究，但是心裡還是有一點遺憾，似乎和原本實驗室的同學與學弟妹們愈走愈遠。

不只是回到了中央，居然也順便回到了大學母校，還回到了當初拉我一把的黃衍介教授實驗室，這真的是緣份。想當初還是大學生時打掃的無塵室，居然在大學畢業了幾年後，變成了我的主力製程實驗室。原本以為不會再見面的一群碩士班新生，居然又在這間實驗室各據一方各自為了博士學位努力。

生活至此我已經很習慣奔波於台北到新竹的兩條高速公路之間，台北、中壢與新竹，輪流在一天中交替出現在眼前。甚至美國東西兩岸的研討會場也成了每年必要的朝聖地。而在美國，緣份更是神奇，居然遇到了應該要在台灣才能遇見的舊識們，有離開大學母校後以為不會再見面的學長姐，也有離開台大後就音訊全無的學弟們。

研究成果在博士班生涯後段不斷開出，自己也意識到了離畢業愈來愈近，可是卻又覺得只差一小步的距離怎麼會耗掉我如此多的力氣。幸虧此時有了清大楊士禮教授與他那群學生的鼎力協助，加上張正陽教授雪中送炭的儀器贊助與論文指導，終於是跨過了這小小的一段距離。這一小段路，非常感謝親愛的家人們始終如一的照顧，父母經濟上的支援，太太辭去工作和我每日抱著儀器爬上爬下，女兒也在此時降臨給予我精神上的安慰。

一路跌跌撞撞走來除了家人以外，我也靠著不少人的打氣與幫助才能撐下來，因陣容龐大請恕我無法一一將姓名列出。感謝張正陽老師實驗室的同學和學弟妹們、陳彥宏老師實驗室的學弟妹們、伍茂仁老師實驗室的學弟妹們、清大黃衍介老師與楊士禮老師實驗室的同學與學弟妹們、台大彭隆瀚老師實驗室的同學與學弟妹們和中研院原分所孔慶昌研究員實驗室的助理們。

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Chapter 1 Introduction

In the past two decades the science and technology for the quasi-phase-matching (QPM) techniques in optical parametric nonlinear wavelengths conversions have progressed tremendously. Periodically poled lithium niobate (PPLN) has been proven as a practical QPM material for high-efficiency nonlinear frequency conversion due to its large effective nonlinear coefficient and free of walk-off problems in nonlinear wavelength conversion.

The concept of electro-optic (EO) QPM has been proposed in 2000 [1]. EO QPM utilizes periodically inverted ferroelectric domains to modulate EO coefficients in third-rank tensors. EO effect along with spatially periodic sign reversal EO coefficients which came from anti-parallel ferroelectric domains in PPLN having a domain width equivalent to a half-wave plate can result in the swing of the polarization of a wave incident to it and increase its azimuth angles with respect to the crystal optic axis in proportion to the domain numbers. Thus it acts as an electric controlled folded Šolc-filter-like polarization rotator. A wave has to satisfy a specific EO QPM condition to perform such a polarization-mode (PM) conversion process in EO PPLN. An electro-optic wavelength filter can be implemented using such an EO QPM PM converter when a pair of polarizers are used. Furthermore, the EO PPLN can be placed in a laser cavity having a polarization-dependent loss element such as a polarization-dependent laser gain medium, for example Nd:YVO_4, Nd:$LiNbO_3$ [2], and so on, to work as a Q-switch just like a conventional Pockels' cell to introduce additional laser cavity loss via switching the laser polarization between high- and low-loss laser modes.

Base on the proposed EO QPM concept various ideas and applications such as polarization mode converters [1], wavelength filters [3][4], and active laser Q-switches [5] in PPLN crystals have been demonstrated.

In 2000 Lu and others [1] first proposed such an EO effect of PPLN an its applications to the polarization-mode conversion. Besides the nonlinear coefficient, other physical coefficients in the form of third-rank tensors are also modulated periodically due to the periodic ferroelectric domains in PPLN including the piezoelectric coefficient and EO coefficient. As a consequence, it is natural to ask the question: what will happen if the EO coefficient modulation is considered? In seeking the answer of the question, they demonstrated that if an electric field is applied along the crystallographic *y*-axis of PPLN, coupling between the extraordinary wave and ordinary wave is established. Under such an EO QPM condition for PM conversion, the polarization state of light propagating in the crystal could be modulated by the applied electric field as shown in Fig. 1.1 [1]. The basic theoretical model can be constructed with the coupled-mode theory which is similar to the analysis made for Šolc filter in text book but with EO induced perturbation. Then an active Šolc filter like PM converter can be realized via such an EO PPLN. Such a PM converter can be employed to act as a wavelength filter by installing a pair of polarizer and analyzer respectively in two ends of the PM converter, and it did be realized in July 2003 by X. Chen and others [3,4].

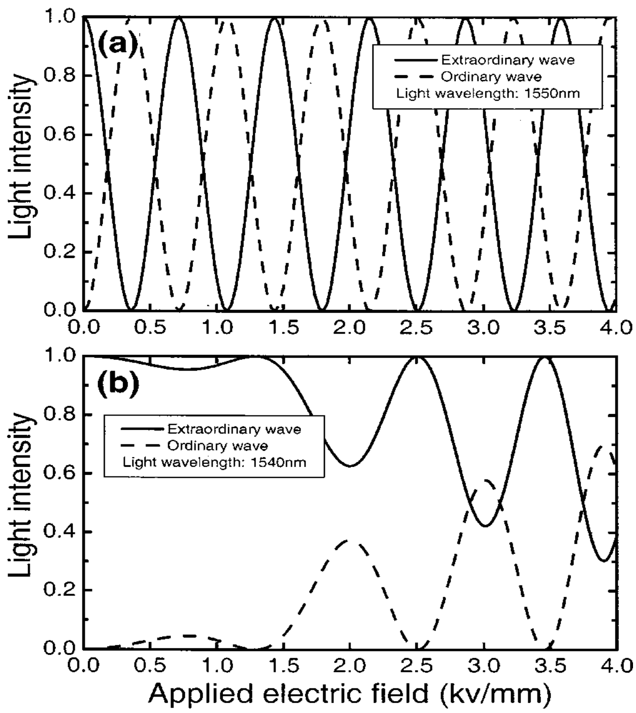


Fig. 1.1 [1] Power exchange relations between the extraordinary wave and the ordinary wave when (a) the QPM condition is satisfied and (b) the QPM condition is not satisfied. The solid line and the dashed line correspond to the intensity-varying curve of the two waves along the propagation direction, respectively.

In 2003, X. Chen and others have experimentally demonstrated the EO PPLN wavelength filter. They were inspired from that a Šolc filter is well known as a polarization interference filter made of birefringent crystals, and an electrically active Šolc filter can be fabricated by application of a periodic array of equal and opposite voltages to an array of thin platelet of $LiTaO_3$ [6]. They compared the EO PPLN wavelength filter with traditional Šolc filters and showed the EO PPLN Šolc filter takes advantage of one-chip integration in lithium niobate and a simpler electrode structure. In particular, they observed the amplitude modulation of the transmission power by applying an electric voltage along the transverse direction of the PPLN as shown in Fig. 1.2 [4]. In addition, the wavelength of the transmission light can be tuned by the temperature over a broad range as shown is Fig. 1.3 [4]. Then one may understand that the EO PPLN wavelength filter is basically the same with an EO PPLN polarization rotator plus a pair of polarizers. QPM wavelength can be tuned since lithium niobate changes its birefringence and dispersion with temperature.

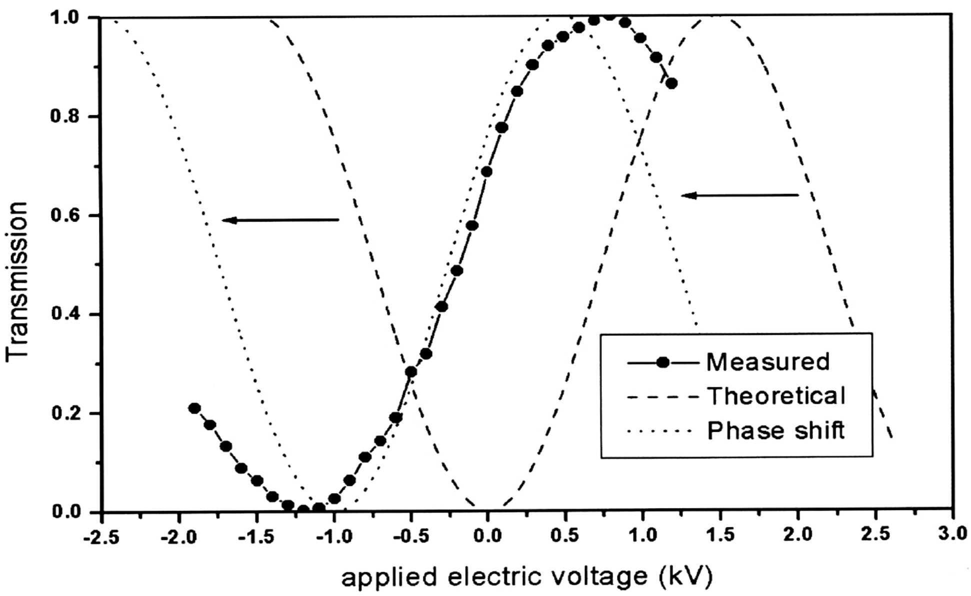


Fig. 1.2 [4] Experimental measurement of the normalized transmission of the Šolc filter (20.8 mm) as a function of the applied voltage for a given wavelength of 1529.80 nm. The solid curve is the experimental measurement, the dashed curve represents the theoretical values calculated from the Sellmeier equation, and the dotted curve is phase shifted from the dashed curve for comparison with the experiment.

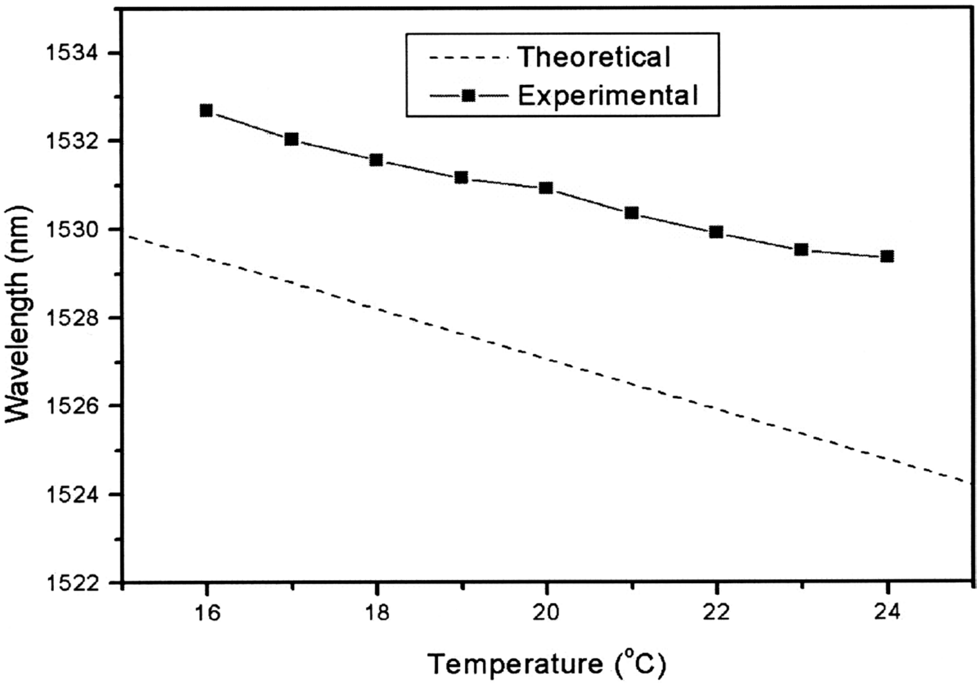


Fig. 1.3 [4] Experimental measurement of the central wavelength of the Šolc filter (20.8 mm) as a function of the temperature without the applied external electric voltage. The dashed line represents the theoretical values calculated from the Sellmeier equation, and the solid curve is the experimental measurement.

In April 2003 Y. H. Chen and Y. C. Huang [5] proposed another practical application based on the EO PPLN technique, EO PPLN as an active laser Q-switch, which opens a new research topic. They revealed that an EO PPLN can function as an active laser Q-switch in a laser having polarization-dependent loss element such as a Brewster plate or a polarization-dependent laser gain medium such as a Nd:YVO_4 crystal. Chen's research also demonstrated that the laser Q-switch made from EO PPLN can be driven in low voltage as shown in Fig. 1.4 [5] with fast modulation speed. They found the half-wave voltage, voltage required for rotating a TM/TE-polarized input wave by 90°, of an EO PPLN Q-switch is only about half of that for an EO Pockels' cell made of single-domain lithium niobate single domain crystal under the same dimensions and operation wavelength. The normalized half-wave voltage they measured was 0.36 V × d (μm)/ L (cm) at wavelength 1064 nm. In this dissertation the further progress of this technique to realize a monolithic solid state internal Q-switched laser in EO Nd:MgO:PPLN will be reported.

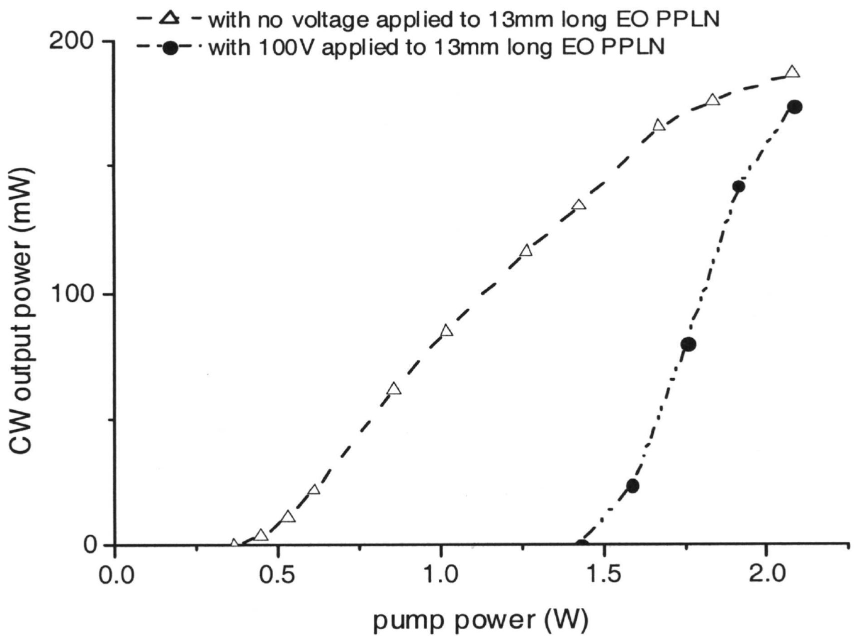


Fig. 1.4 [5] CW performance of the $Nd:YVO_4$ laser with (solid curve) and without (dashed curve) a 100-V voltage applied to the EO PPLN. With the 100-V voltage, the laser threshold was increased from 400 mW to 1.4 W because of the polarization-rotation effect from the EO PPLN crystal.

Since PPLN can functions as an QPM EO Q-switch, and an QPM frequency converter, Chen and others in 2005 have demonstrated a monolithic PPLN crystal integrating both functions simultaneously to construct an intracavity optical parametric oscillator (IOPO) [7]. Their research work demonstrated first success in integrating a QPM wavelength converter and an EO QPM PM converter in a monolithic QPM crystal to achieve a useful device. In this way, an integrated QPM device can be versatile in function by ingenious integration of its electro-optic and wavelength-conversion advantages in a monolithic crystal substrate and can be readily implemented through nowadays mature micro-fabrication techniques. In this work they reported the performance of a monolithic QPM device constructed for simultaneous laser Q switching and an optical parametric gain medium in a diode-pumped $Nd:YVO_4$ laser as shown in Fig. 1.5 [10] achieve a pump Q-switched IOPO at near the 1.55-μm eye-safe wavelength. The IOPO has been studied since early 1970s [8][9], but it did not draw much attention until recently perhaps owing to the difficulty in optimizing both laser Q switch and a nonlinear optical material in a single laser cavity. An optical parametric oscillator internal to a Q-switched laser can take the advantage of the high intracavity pump power for efficient wavelength conversion. Then this laser can be a diode-pumped dual-wavelength, fundamental and signal, Q-switched laser which can be a high power and high repetition rate optical telecommunication light source and an eye-safe laser as well.

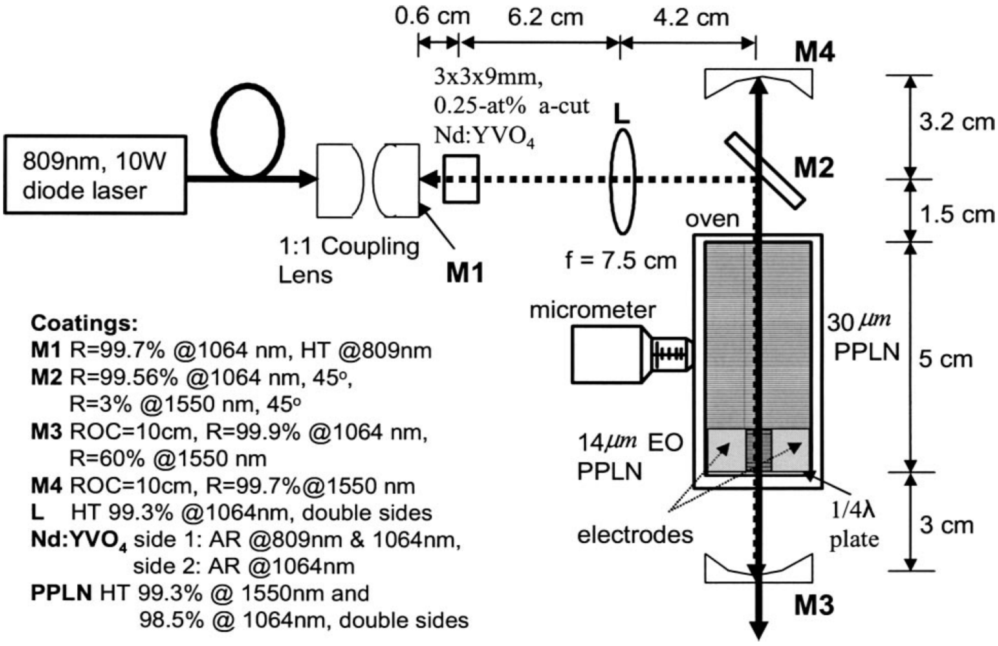


Fig. 1.5 [10] The schematic of a 9-W diode-pumped Nd:YVO4 Q-switched laser comprising a 5-cm long, 30-μm-period PPLN wavelength converter cascaded to the E-O PPLN Pockels cell. R: reflectance; M1, M2, M3, and M4 are cavity mirrors.

In 2004 it was active narrowband multiple wavelengths filter [11] that was theoretical proposed as an important research progress based on EO PPLN. The narrowband multiple wavelengths filter was constructed in aperiodic optical superlattice (AOS) which can be realized in an EO aperiodically poled lithium niobate (APLN) crystal. The binary sequence of the domains in AOS is optimized to realize the prescribed multiple wavelengths filtering using the simulated annealing (SA) algorithm [44][14]. The potential application for such a multi-wavelength filter is at the optical communication system such as the dense wavelength division multiplexing (DWDM) multiplexer/de-multiplexer (MUX/DeMUX). In this dissertation a new iterative design scheme and the first experimental demonstration of an 8-channel DWDM filter based on EO APLN will be reported in chapter 4. The derivation of the idea for this EO AOS was inspired by the AOS technique initially developed for performing simultaneously QPM multi-wavelength conversion

[12][13][14][15][16]. Those research works in aperiodic optical superlattice with SA algorithm [14,15] and Damman grating [16] successfully demonstrated the capability to suppress unwanted sideband signal generations. In this dissertation the demonstration of the construction of a *single* aperiodic QPM-domain structure for simultaneously phase-matching electro-optic and nonlinear-optic wave-energy couplings in $LiNbO_3$ to achieve the high-efficiency multi-channel operation of dual optical device functions will be reported in chapter 5. From the nonlinear optics theory, PPLN wavelength conversion efficiency is proportional to the input fundamental power and the interaction length. Thus such nonlinear optics devices were favorably placed in laser cavities to achieve much higher conversion efficiencies just as the monolithically cascade EO PPLN Q-switch and OPO device aforementioned [7,10]. However the conversion efficiencies could be still limited by the diffraction and clipping losses owing to the small aperture of PPLN bulk crystals (~0.5-1 mm) comparing with other conventional birefringence nonlinear crystals. Then the system alignment became more and more critical as intracavity PPLN length increases. Since the AOS in PPLN has been used to generate multiple reciprocal vectors for QPM multi-wavelength conversion or multi-wavelength PM conversion, another novel device of utilizing AOS technique to hybridize reciprocal vectors in a *single* APLN for simultaneously performing QPM wavelength conversion and PM conversion has been also studied in this dissertation. The design and experimental demonstration of an EO Q-switched SHG laser using a single APLN crystal will be presented in chapter 5. In this work, a diode-pumped $Nd:YVO_4$ laser radiating in 1342 nm was Q-switched and frequency-doubled by a 2-cm long APLN chip having single AOS structure.

The development of OPOs has gained rapid progress with the aid of QPM techniques as well [17][18][19]. OPOs have been one of promising techniques in producing compact, broadly tunable coherent sources of radiation. Generally, broad tuning in wavelength can conveniently be done via changing the temperature of the optical parametric gain medium, which is slow. For many applications such as the spectroscopy, a narrow-line light source is demanded. There are several techniques to produce a narrow-line light source in QPM PPLN OPO [20][21][22]. In this dissertation an gain-enhanced and spectral-narrowed OPO has been proposed and demonstrated based on a monolithic PPLN integrating electro-optically active (EOA) PM converters with an optical parametric gain medium pumped by an actively Q-switched Nd:YAG laser. This work provides a preliminary research study for obtaining a reliable model for further design to achieve high efficiency, broad tuning, and spectral manipulation in a compact OPO device with fast tuning capability.

Chapter 2 Principles of EO PPLN and Its Properties

The basic concept and theory of using EO PPLN as a Šolc-filter-like polarization rotator were proposed by Lu and others [1] in 2000. Then X. Chen and others demonstrated this concept experimentally [4]. In 2003 Y. H. Chen conducted another experiment [5] to demonstrate an idea of using an EO PPLN as a low-voltage laser Q-switch. Based on these demonstrated concept and experiments, this research works have further devoted developing more advanced devices and applications which will be detailed in the following chapters

In this chapter the basic theory of a folded Šolc filter will be briefly introduced first, then the operational principle of an EO QPM device will be analyzed.

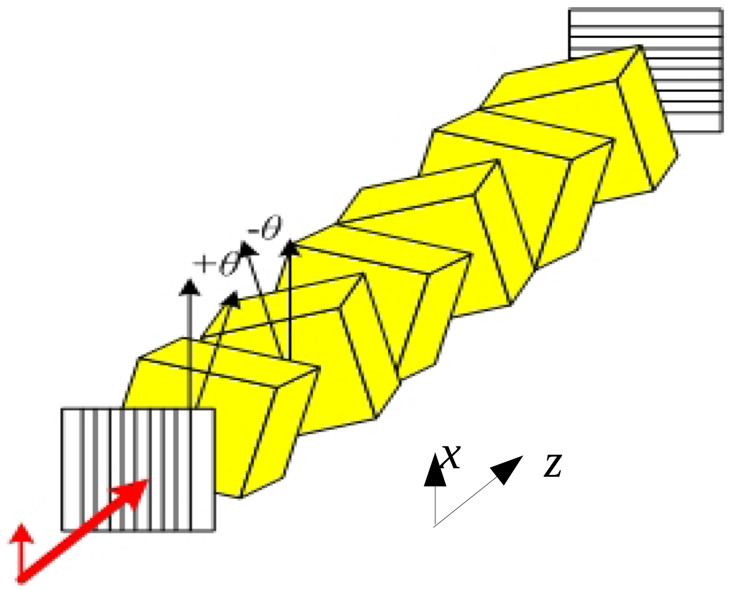


Fig. 2.1 Sketch of the geometric arrangement of a folded Šolc filter. The retardation plate is with azimuth angle θ with respect to x axis.

Folded Šolc filter is one of the two basic types of Šolc filters. Fig. 2.1 shows the sketch of the geometric arrangement of the filter. Transmission axis of the front polarizer is parallel to the x axis, and the rear one is parallel to the y axis. One may model the folded Šolc filter with Jones matrix calculations or coupled-mode equations. With the first calculation method [23], the overall Jones matrix for these N wave retardation plates is given by

$$M=[R(\theta)W_0 R(-\theta)R(-\theta)W_0 R(\theta)]^m \quad , \tag{2-1}$$

where m is the assumed number of plates which is an even number, $N = 2m$,

$$R(\theta)=\begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix} \tag{2-2}$$

is the rotation matrix, and

$$W_0=e^{-i\phi}\begin{pmatrix} e^{-i\Gamma/2} & 0 \\ 0 & e^{i\Gamma/2} \end{pmatrix} \tag{2-3}$$

is the Jones matrix for the retardation plate. The retardation plate is with azimuth angle θ with respect to x axis and phase retardation Γ. Assume that each plate is characterized by refractive indices n_e and n_o for extraordinary wave and ordinary wave respectively at certain wavelength λ and by thickness d, then the phase retardation at a given wavelength is

$$\Gamma=2\frac{\pi}{\lambda}(n_e-n_o)d \quad . \tag{2-4}$$

One may carry out the matrix multiplication, resulting in

$$M=\begin{pmatrix} A & B \\ C & D \end{pmatrix} \tag{2-5}$$

with

$$A=[\cos(\frac{1}{2}\Gamma)-i\cos(2\theta)\sin(\frac{1}{2}\Gamma)]^2+\sin^2(2\theta)\sin^2(\frac{1}{2}\Gamma) \quad ,$$

$$B=\sin(4\theta)\sin^2(\frac{1}{2}\Gamma) \quad ,$$

$$C=-B \quad ,$$

$$D=[\cos(\frac{1}{2}\Gamma)+i\cos(2\theta)\sin(\frac{1}{2}\Gamma)]^2+\sin^2(2\theta)\sin^2(\frac{1}{2}\Gamma) \quad . \tag{2-6}$$

Since M is unimodular, it can be simplified to

$$M=\begin{pmatrix} \dfrac{A\sin mK\Lambda-\sin(m-1)K\Lambda}{\sin K\Lambda} & B\dfrac{\sin mK\Lambda}{\sin K\Lambda} \\ C\dfrac{\sin mK\Lambda}{\sin K\Lambda} & \dfrac{D\sin mK\Lambda-\sin(m-1)K\Lambda}{\sin K\Lambda} \end{pmatrix} \tag{2-7}$$

with

$$K\Lambda=\cos^{-1}[\frac{1}{2}(A+D)] \quad . \tag{2-8}$$

The transmitted wave, E', and incident wave, E, can be related by $E' = P_y M P_x E$ where P_y and P_x represent the Jones matrix of crossed polarizers for having transmission axes at y and x respectively. The transmitted wave from this filter is polarized in y direction, and its field amplitude is given by $E'_y = M_{21}E_x$. Assume that the incident wave is linearly polarized in x direction, then the transmittance of this filter is given by

$$T=|M_{21}|^2=\left|\sin 4\theta\sin^2\frac{1}{2}\Gamma\frac{\sin mK\Lambda}{\sin K\Lambda}\right|^2 \tag{2-9}$$

with

$$\cos K\Lambda=1-2\cos^2 2\theta\sin^2\frac{1}{2}\Gamma \quad . \tag{2-10}$$

When the phase retardation of each plate is $\Gamma = \pi, 3\pi, 5\pi, \ldots$, the transmittance becomes $T = \sin^2 2N\theta$. It means that each plate is a half-wave plate. When th azimuth angle θ is

$$\theta=\frac{\pi}{4N} \quad , \tag{2-11}$$

this transmittance becomes 100%.

After the light wave passing through the front polarizer, it is linearly polarized in the x with zero azimuth angle. But the optic axis of the first plate is at the azimuth angle θ, the transmitted wave will then be linearly polarized at 2θ after transmitting this plate. The optic axis of the second plate is at the azimuth angle $-\theta$, which then results in an angle of 3θ with respect to the polarization of light incident on it. Thus the total polarization state will oriented at -4θ with respect to its original state of incidence as shown in Fig. 2.2. In short, the polarization rotation angle of the wave is 4 times of the half-wave plate azimuth angle θ after passing through a pair of plates. The final azimuth angle of the wave after N plates is thus $2N\theta$. If the $2N\theta$ equals 90°, then the wave can 100% transmit the rear polarizer.

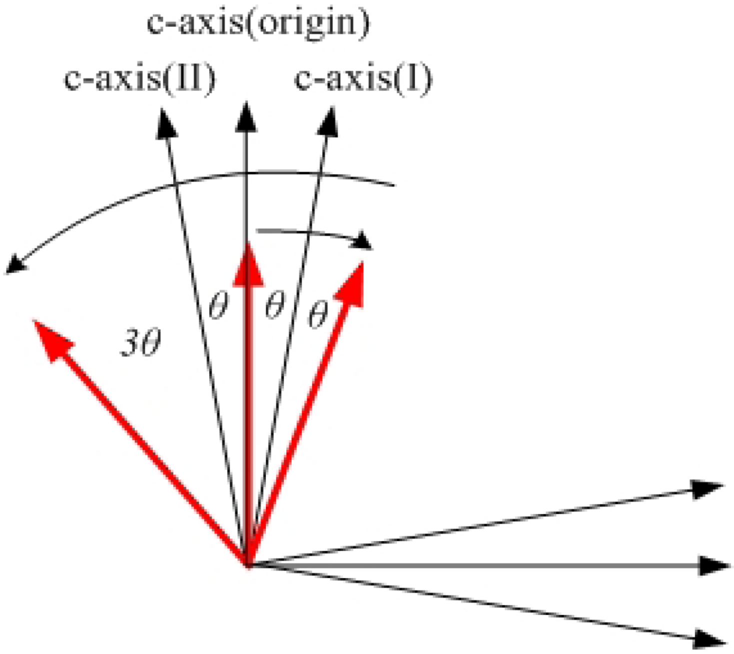


Fig. 2.2 In the folded Šolc filter the polarization rotation angle of the wave is 4 times of the half-wave plate azimuth angle θ after passing through a pair of plates.

Folded Šolc filter can also be studied via coupled-mode theory which is helpful to understand the detail physical mechanism of an EO QPM structure. The coupled modes in a folded Šolc filter are assumed to propagate in the same direction, i.e., codirectional coupling.

The studied substrate, lithium niobate ($LiNbO_3$), is a ferroelectric crystal with the symmetry of class 3m. It is easy to demonstrate that all elements of the electro-optic tensor of the crystal have different signs in different domains. The impermeability tensor is defined as

$$\eta_{ij} = \varepsilon_0 (\varepsilon^{-1})_{ij} \quad , \tag{2-12}$$

where ε^{-1} is the inverse of the permittivity tensor ε, ε^0 is the permittivity of vacuum, and i, j (1, 2, 3) are repeated the cartesian-component indices. The index ellipsoid can be expressed by

$$\eta_{ij}\xi_i\xi_j=1 \quad , \tag{2-13}$$

where ξ_1, ξ_2, ξ_3 are the coordinates of an arbitrary point in the new coordinate system. The application of an electric field *E* will result in a slightly change in the impermeability tensor

$$\Delta\eta_{ij}\equiv\eta_{ij}(E)-\eta_{ij}(0)\approx\gamma_{ijk}E_k \quad , \tag{2-14}$$

where γ is the linear (Pockels) EO coefficient. The EO coefficients tensor of $LiNbO_3$ is in the form

$$\gamma_{ijk}=\begin{bmatrix} 0 & -\gamma_{22} & \gamma_{13} \\ 0 & \gamma_{22} & \gamma_{13} \\ 0 & 0 & \gamma_{33} \\ 0 & \gamma_{51} & 0 \\ \gamma_{51} & 0 & 0 \\ -\gamma_{22} & 0 & 0 \end{bmatrix} \quad . \tag{2-15}$$

The geometrical arrangement for studying such an EO effect is shown in Fig. 2.3. In the presence of an external electric field along the crystallographic *y* axis, $\boldsymbol{E_y}$, the impermeability tensor is given by

$$\eta_{ij}(E)=\begin{bmatrix} \frac{1}{n_o^2}-\gamma_{22}E & 0 & 0 \\ 0 & \frac{1}{n_o^2}+\gamma_{22}E & \gamma_{51}E \\ 0 & \gamma_{51}E & \frac{1}{n_e^2} \end{bmatrix} \quad , \tag{2-16}$$

therefor the index ellipsoid deforms to

$$\left(\frac{1}{n_o^2}+\gamma_{22}E\right)x^2+\left(\frac{1}{n_o^2}-\gamma_{22}E\right)y^2+\frac{1}{n_e^2}z^2+2yz\gamma_{51}E=1 \quad , \tag{2-17}$$

and a rotation of the ellipsoid in the *yz* plane is possible to diagonalize its corresponding impermeability tensor. The transformation from *y*, *z* to *y'*, *z'* is given by

$$\begin{bmatrix} y \\ z \end{bmatrix}=\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}\begin{bmatrix} y' \\ z' \end{bmatrix} \quad . \tag{2-18}$$

The coefficient of the *y'z'* term after transformation should be vanished, and for $LiNbO_3$

$$\left(\frac{1}{n_e^2}-\frac{1}{n_o^2}\right) \gg \gamma_{22} E \tag{2-19}$$

which leads to the the rotation of the index ellipsoid a small angle,

$$\theta \approx \frac{\gamma_{51} E}{\left(\frac{1}{n_e^2}-\frac{1}{n_o^2}\right)} \quad , \tag{2-20}$$

about the x axis.

With the consideration of the periodic sign change of the Pockels' coefficients, the dielectric constant of PPLN with $\boldsymbol{E_y}$ can thus be written as

$$\varepsilon = \varepsilon(0) + \Delta\varepsilon\, s(x) \quad , \tag{2-21}$$

where

$$\varepsilon(0) = \varepsilon_0 \begin{bmatrix} n_o^2 & 0 & 0 \\ 0 & n_o^2 & 0 \\ 0 & 0 & n_e^2 \end{bmatrix} \tag{2-22}$$

is the original dielectric tensor without $\boldsymbol{E_y}$,

$$\varepsilon = \varepsilon_0 [\eta_{ij}(E)]^{-1} \approx \varepsilon_0 n_o^2 n_e^2 \begin{bmatrix} \frac{1}{n_e^2} & 0 & 0 \\ 0 & \frac{1}{n_e^2} & \gamma_{51} E \\ 0 & \gamma_{51} E & \frac{1}{n_o^2} \end{bmatrix} \quad , \tag{2-23}$$

is the permittivity tensor, and

$$\Delta\varepsilon = -\varepsilon_0 \gamma_{51} E\, n_o^2 n_e^2 \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \tag{2-24}$$

is the change of the dielectric tensor. In PPLN, a sign function

$$s(x) = \pm 1 \tag{2-25}$$

which changes its values (+1 or -1) along with ferroelectric domain polarities should be included. The sign function can also be expressed by a Fourier's series as

$$s(x)=\sum_{m\neq 0} G_m e^{iK_m x}, |G_m|=\frac{2}{m\pi}\sin(m\pi D) \quad , \tag{2-26}$$

where $D \equiv l\,/\,\Lambda$ is the duty cycle with l the length of the positive domain block, and Λ is the modulation period.

The change of the dielectric tensor could be viewed as a perturbation to the dielectric structure, thus it can be expressed by the coupled-mode equations of codirectional coupling [23]

$$\begin{aligned}\frac{dA_o}{dx}&=-i\kappa A_e e^{i\Delta\beta x}\\ \frac{dA_e}{dx}&=-i\kappa^* A_o e^{-i\Delta\beta x}\end{aligned} \quad , \tag{2-27}$$

having the two modes satisfying the conservation of energy, which requires that

$$\frac{d}{dz}(|A_o|^2+|A_e|^2)=0 \quad . \tag{2-28}$$

where

$$\Delta\beta=(\beta_o-\beta_e)-K_m,\; K_m=\frac{2\pi m}{\Lambda} \tag{2-29}$$

is the phase mismatch between the two modes, A_o and A_e are the amplitudes of the ordinary wave and the extraordinary wave, respectively, β_o and β_e are the corresponding wave vectors. $K_m \equiv 2\pi/\Lambda$ is the m^{th} order reciprocal vector, Λ is equal to twice domain thickness if the duty cycle D is 50%. And the coupling coefficient κ is defined as [1][24]

$$\kappa=\frac{\omega^2\mu_0}{2\sqrt{\beta_o\beta_e}}\hat{e}_o\cdot\Delta\varepsilon(x)\hat{e}_e=\frac{\omega}{c}\frac{n_o{}^2 n_e{}^2\gamma_{51}E}{\sqrt{n_o n_e}}\frac{(\sin m\pi D)}{m\pi},(m=1,3,5\ldots) \quad , \tag{2-30}$$

where ω is the angular frequency of modes, μ_0 is the permeability of vacuum, and $\hat{e}_o$ and $\hat{e}_e$ are the unit polarization vectors of the two normal modes. Assuming the incident light is an extraordinary wave, the initial condition at x=0 is $A_e(0)$=1, $A_o(0)$=0. We may solve these equations with following steps. By decoupling the coupled equations, one can easily obtain as

$$\frac{d^2 A_o}{dx^2} = -|\kappa|^2 A_o + i\Delta\beta \frac{dA_o}{dx} \quad , \tag{2-31}$$

and its basis of general solution is

$$A_o(x) = e^{i\Delta\beta\frac{x}{2}}[P_1\cos(rx) + P_2\sin(rx)] \quad . \tag{2-32}$$

Since $A_o(0)=0$, $P_1=0$ is found. The basis of general solution can be simplified as

$$A_o(x) = P_2 e^{i\Delta\beta\frac{x}{2}}\sin(rx) \quad , \tag{2-33}$$

where

$$r^2 = |\kappa|^2 + \left(\frac{\Delta\beta}{2}\right)^2 \quad . \tag{2-34}$$

Then introduce Eq. (2-33) into Eq. (2-27), the equation becomes

$$P_2[i\frac{\Delta\beta}{2}e^{i\Delta\beta\frac{x}{2}}\sin(rx) + e^{i\Delta\beta\frac{x}{2}} r\cos(rx)] = -i\kappa A_e e^{i\Delta\beta x} \quad . \tag{2-35}$$

Using the initial condition $A_e(0)=1$ then $P_2 = -i\frac{\kappa}{r}$ is got. From Eq. (2-33)

$$A_o(x) = -i\frac{\kappa}{r}e^{i\Delta\beta\frac{x}{2}}\sin(rx) \tag{2-36}$$

and

$$A_o^*(x) = i\frac{\kappa^*}{r}e^{-i\Delta\beta\frac{x}{2}}\sin(rx) \tag{2-37}$$

are the solutions, and

$$|A_o(x)|^2 = |\kappa|^2\frac{\sin^2(rx)}{r^2} \tag{2-38}$$

can be found. The maximum value of $A_o(x)$ occurs at

$$\Delta\beta = 0 = (\beta_o - \beta_e) - K_m \tag{2-39}$$

which is the QPM condition for such a PM converter, and

$$|\kappa|x=\frac{q\pi}{2},\ q=1,3,5,\ldots \tag{2-40}$$

which is called dynamical condition [1].

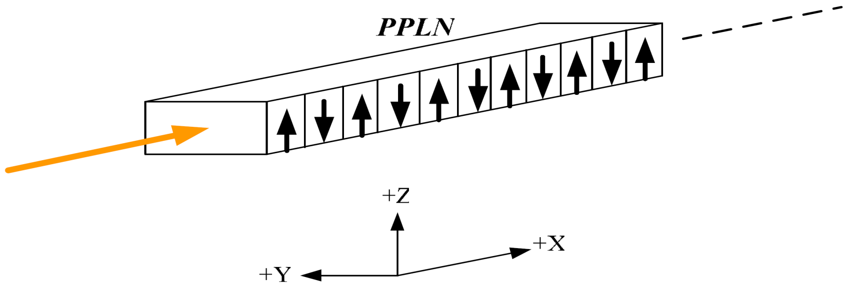


Fig 2.3 The geometrical arrangement for studying EO effects in PPLN.

In the general case, for a given static field, the transmittance of this filter is thus obtained as

$$T=\left|\frac{A_o(x)}{A_e(0)}\right|^2=|\kappa|^2\frac{\sin^2(rx)}{r^2}\quad , \tag{2-41}$$

If the field is not strong enough or the crystal length is too short such that $rx<<\pi/2$, the coupling between the extraordinary wave and ordinary wave is weak, and the weak coupling approximation, $A_e(x)=A_e(0)=1$, $Db>>|k|^2$ can be used. Thus the power conversion efficiency obtained in Eq. (2-41) now becomes

$$T=\left|\frac{A_o(x)}{A_e(0)}\right|^2=\kappa^2x^2\left(\frac{\sin(\Delta\beta\frac{x}{2})}{\Delta\beta\frac{x}{2}}\right)^2=\kappa^2x^2 sinc^2(\Delta\beta\frac{x}{2})\quad . \tag{2-42}$$

This expression is similar to that of the QPM frequency conversion efficiency, thus the maximum conversion efficiency is achieved as $\Delta\beta=0=(\beta_o-\beta_e)-K_m$ as well. This condition occurs when the mismatch wave vector of this coupling process is compensated by the reciprocal vector with the QPM structure and we can name it as the QPM condition which is similar to QPM wavelength conversion. The coherent length can be defined as [1]

$$L_c = 2\frac{\pi}{\lambda(n_o - n_e)} \tag{2-43}$$

for a given wavelength. The coupling efficiency lowered by the unsatisfactory QPM condition may be further enhanced by a proper increase of the applied electric field as revealed in Eq. (2-41). This is similar to the enhancement with the increase of the input pump power in QPM nonlinear wavelength conversion process.

From the dynamical condition Eq. (2-40) we can readily find the electric field,

$$E = \frac{c}{\omega}\frac{q}{n_o^{\frac{3}{2}} n_e^{\frac{3}{2}}}\frac{m\pi^2}{(1-\cos m\pi)}\frac{1}{L}, q = 1,3,5,\ldots, m = 1,3,5,\ldots \quad , \tag{2-44}$$

to reach the 100% conversion with the filter for a given crystal length L under $\Delta\beta = 0$.

In addition to the studies of periodic QPM grating materials like PPLN, recently much attention has been given to the achievement of non-uniform grating designs in QPM materials using Fibonacci optical superlattice (FOS) [12], phase-reversal sequence (PRS) [13], and aperiodic optical superlattice (AOS) [14] techniques. These techniques can be utilized to create spatially modulated domain lattices of QPM material which are capable of providing a number of reciprocal vectors for the simultaneous compensation of a corresponding number of wave-vector mismatches, thereby allowing for efficient multiple wave-energy conversions.

In this study, several novel EO APLN devices are also proposed and demonstrated. The principles an EO APLN are discussed as follows.

The uniform grating structure in PPLN crystal provides only one reciprocal vector and therefore can only phase-match to one designated wave coupling process. An aperiodic domain structure in a QPM material can thus be regarded as a structure composing of more than one reciprocal vector so that simultaneous compensation of wave-vector mismatches of a plurality of wave-energy coupling processes is possible. One of the unique advantages of a QPM crystal is its high flexibility in the domain patterning and engineering. In order to compute and to optimize the domain structure for manipulating the relative efficiency among the multiple coupling processes, a iterative solution of the EO QPM process has to be developed first. Consider now the two normal modes described by Eq. (2-27) codirectionally propagate in an EO APLN crystal. Assume that the APLN is constructed by N blocks with each block a thickness of Δx as shown in Fig. 2.4 and a polarity of either +1 or -1. Let $x_j = j\Delta x$ represent the position of the j^{th} block of the APLN for j = 0, 1, 2, …, (N-1), the general solution of Eq. (2-27) can be expressed as

$$\begin{aligned} A_o(x_{j+1}) &= e^{i\frac{\Delta\beta}{2}x_{j+1}}[P_1(x_j)\cos rx_{j+1} + P_2(x_j)\sin rx_{j+1}] \\ A_e(x_{j+1}) &= e^{-i\frac{\Delta\beta}{2}x_{j+1}}\{P_1(x_j)[\frac{-\Delta\beta}{2\kappa(x_{j+1})}\cos rx_{j+1} - \frac{ir}{\kappa(x_{j+1})}\sin rx_{j+1}] \\ &\quad + P_2(x_j)[\frac{-\Delta\beta}{2\kappa(x_{j+1})}\sin rx_{j+1} + \frac{ir}{\kappa(x_{j+1})}]\} \\ j &= 0, 1, 2, \ldots (N-1) \end{aligned} \quad , \qquad (2\text{-}45)$$

where

$$\begin{aligned} P_1(x) &= \sin rx(i\frac{\kappa(x)}{r}A_e(x)e^{i\frac{\Delta\beta}{2}x}) + [i\frac{\Delta\beta}{2r}\sin rx + \cos rx]A_o(x)e^{-i\frac{\Delta\beta}{2}x} \\ P_2(x) &= \cos rx(-i\frac{\kappa(x)}{r}A_e(x)e^{i\frac{\Delta\beta}{2}x}) - [i\frac{\Delta\beta}{2r}\cos rx - \sin rx]A_o(x)e^{-i\frac{\Delta\beta}{2}x} \end{aligned} \quad . \qquad (2\text{-}46)$$

Assume that it is the ordinary wave (i.e., TE mode) incident in the EO APLN, and the initial conditions are $A_e(0)$=0 and $A_o(0)$=1. Under this condition, the conversion efficiency for the mode extraordinary wave (i.e., TM mode) of the EO APLN PM converter that is defined by

$$T = \eta_{EO} = \left|\frac{A_e(x_N)}{A_o(0)}\right|^2 \qquad (2\text{-}47)$$

can thus be iteratively calculated from Eq. (2-41). The transmittance obviously depends on the domain polarity composition of the N blocks. Now these solutions and transmittance can be used to compute binary domain structures in EO APLN. With the aid of optimization computing algorithm, an optimum domain sequence would be obtained.

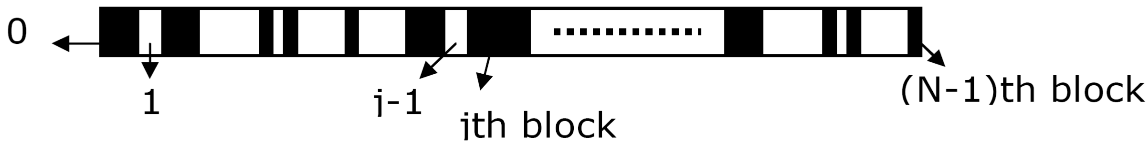


Fig. 2.4 Binary sequence arrangement of an APLN device.

Based on these principles several novel and practical devices have been proposed and demonstrated in this research work. Still other relevant theories such as the nonlinear wavelength conversion and laser Q-switching will be used in the following chapters for characterizing the devices proposed.

Chapter 3 EO PPLN as An Intracavity Q-switch for Laser

Introduction

$LiNbO_3$ is a popular optical substrate for the construction of various optical and photonic devices such as wave plates, modulators, scanners, and so on. With the aid of the impurity doping or metal indiffusion technique, $LiNbO_3$ can also become a promising host material for various laser ions, such as Nd^{3+}, Er^{3+}, and Yb^{3+}. Thus, laser ion-doped or diffused $LiNbO_3$ has now become a medium of particular interest for the implementation of monolithic self-functioned laser devices such as self-frequency-doubled and internal-Q-switched lasers. The integration of such a medium with nowadays popular diode-pumping technique could be an attractive scheme for producing a miniature yet efficient multi-function all-solid-state laser system. Nd:MgO:$LiNbO_3$ can be a promising candidate for such purposes, in that the adverse problem of pump induced photorefractive damage found in the early development of Nd:$LiNbO_3$ lasers has been effectively alleviated by doping with ≥ 5 mol. % MgO/ZnO in the crystal [25] and by using a near-infrared pump source [2].

The laser Q-switching technique has proven to be an effective way of producing repetitively high-energy laser pulses with greatly enhanced peak laser power. This technique has also allowed the generation of a peak power greater than several tens of watts in a miniaturized laser. Some typical examples can be found from recent developments in actively and passively Q-switched microchip lasers [26][27]. A class of Q-switched lasers employs a fast intracavity polarization rotator for switching the laser polarization between high- and low-loss laser modes. The two loss modes can be established by using a polarization-dependent loss element such as a Brewster plate or a polarization-dependent laser gain medium. An EO birefringence crystal may change laser polarization state and can be used as a laser Q switch. For example, $LiNbO_3$, KDP, and KTP crystals are usually the crystal materials for Pockels' cells in Q-switched lasers. In particular, Chen and others [5] has incorporated a PPLN crystal in a compact $Nd:YVO_4$ laser to act as a low-voltage EO Q-switch. In this dissertation work, the functionalities of an Nd^{3+} laser medium and an EO PPLN Q-switch were further integrated into a monolithic periodically poled $Nd:MgO:LiNbO_3$ (Nd:MgO:PPLN) crystal to achieve an actively internal-Q-switched laser. In addition to improving system miniaturization, the monolithic integration of laser intracavity elements has the obvious advantages of reducing the cavity loss and the difficulties with laser alignment. Besides, QPM crystals have already been successfully fabricated from advanced $MgO:LiNbO_3$ crystal for the purpose of performing high-efficiency nonlinear frequency conversion with improved laser damage resistance threshold [28]. The use of a Nd:MgO:PPLN crystal to fabricate and demonstrate an internally Q-switched laser in this work may contribute to the realization of a novel internal-Q-switched self-frequency conversion laser in a monolithic $LiNbO_3$ chip.

Q-switch theory

The technique of Q switching allows lasers to generate optical pulses of short duration and high peak power. Suppose a shutter or a loss-modulation element is introduced into the laser cavity. If the losses are high, corresponding to a low cavity Q value, laser oscillation can not occur and cause the population inversion to reach a value far from the threshold population. If the loss mechanism or device is removed or disabled suddenly, the laser sees a gain that greatly exceeds the losses, corresponding to a high cavity Q value, and the s tored energy will be released in the form of a short and intense light pulse. Since this technique switches the laser cavity Q values between a low and a high values, it is named as Q switching.

EO Q-switch makes use of an EO effect, usually the Pockels effect. A Q-switch made of Pockels cell based on this effect mainly consists of a suitable nonlinear crystal such as a KDP or a lithium niobate. Pockels cells are popular Q-switching devices. In a Pockels cell, the crystal birefringence is changed as a function of the applied electric field due to the EO effect, which results in the rotation or change of the polarization state of the wave passing through it. Therefore usually such a Pockels cell has to be collocated with a polarizer in the laser cavity to enable the Q-switching operation.

In the case of a continuously pumped EO Q-switched laser the cavity Q-value is repetitively modulated by the Q-switch, and the laser output consists of a continuous train of Q-switched pulses. During each pulse the inversion population will fall from its initial high value to a final value. The inversion population is then restored to its initial high value by the pumping process before the next Q-switching event.

A number of important features of a Q-switched pulse, such as energy content, peak power, pulse width, rise and fall times, and pulse formation time, can be obtained from the laser rate equations. In all cases of interest the Q-switched pulse duration is so short (usually in the order of ns) that one can neglect the influence of both spontaneous emission rate and optical pumping rate on the temporal pulse formation, i.e., one can neglect both effects in writing the rate equations as follows [29][30]

$$\frac{\partial \phi}{\partial t} = \phi \left(c \sigma n \frac{l}{L} - \frac{\varepsilon}{t_r} \right) \quad , \tag{3-1}$$

$$\frac{\partial n}{\partial t} = -\gamma n \phi \sigma c \quad , \tag{3-2}$$

and

$$\gamma = 1 + \frac{g_2}{g_1} \quad , \tag{3-3}$$

where ϕ is the photon density, c is speed of light in vacuum, σ is the stimulated emission cross section, n is the electron population density, l is the length of active material, L is the length of resonator, ε is the fractional loss per round trip, t_r is the round-trip time, and g_1 and g_2 are the degeneracy of the energy levels of lower (3-1) and upper (3-2) one respectively. The losses in a cavity can be represented by

$$\varepsilon = -\ln R + \delta + \zeta(t) \quad , \tag{3-4}$$

where the first term represents the output coupling losses determined by the mirror reflectivity *R*, the δ contains all the incidental losses, and ζ represents the cavity loss introduced by the Q-switch with the boundary conditions $\zeta(t < 0)=\zeta_{max}$; $\zeta(t > 0)=0$ like a step function.

The equations describing the operation of fast Q-switched lasers involves the the simultaneous solution of two coupled differential equations. One is temporal evolution of the internal photon density in the resonator, Eq. (3-1), and the temporal evolution of the inversion population density in the active medium, Eq. (3-2). The output energy of the Q-switched laser can be expressed as follows:

$$E_{out} = \frac{h\nu A}{2\sigma\gamma} \ln\left(\frac{1}{R}\right) \ln\left(\frac{n_i}{n_f}\right) \quad , \tag{3-5}$$

where $h\nu$ is the laser photon energy and *A* is the effective beam cross sectional area. The initial and final population inversion population densities, n_i and n_f, are related by the transcendental equation

$$n_i - n_f = n_t \ln\left(\frac{n_i}{n_f}\right) \quad , \tag{3-6}$$

where n_t is the inversion population density at threshold, that is,

$$n_t = \frac{1}{2\sigma l}\left(\ln\frac{1}{R} + \delta\right) \quad . \tag{3-7}$$

A new dimensionless variables, *z*, is defined and derived from

$$\frac{n_f}{n_i} = \frac{\delta}{2\sigma n_i l} \equiv \frac{\delta}{2 g_0 l} \equiv \frac{1}{z} \tag{3-8}$$

where g_0 is the unsaturated small-signal gain coefficient. In Ref. [39] an analytical solution has been derived for the optimum reflectivity, given by

$$R_{opt} = \exp\left[-\delta\left(\frac{z - 1 - \ln z}{\ln z}\right)\right] \tag{3-9}$$

and the energy output for the optimized system

$$E_{out}=E_{sc}(z-1-\ln z) \tag{3-10}$$

where

$$E_{sc}=\frac{Ah\nu\delta}{2\sigma\gamma} \tag{3-11}$$

is a scale factor with the dimension of energy. For a four-level laser (Nd:MgO:$LiNbO_3$) γ is one. The FWHM pulse width versus z is obtained from [30]

$$t_p=\frac{t_r}{\delta}\left(\frac{\ln z}{z[1-a(1-\ln a)]}\right) \quad , \quad a=\frac{(z-1)}{(z\ln z)} \quad . \tag{3-12}$$

In the limit of large z, the output energy approaches the total useful energy stored in the gain medium given by

$$E_{st}=\lim_{z\to\infty}E_{out}=\frac{Ah\nu\delta}{2\sigma\gamma}z \quad , \tag{3-13}$$

and one can define an energy extraction efficiency

$$\eta_E=\frac{E_{out}}{E_{st}}=1-\left(\frac{1+\ln z}{z}\right) \quad . \tag{3-14}$$

The peak power of a Q-switch pulse can be calculated from $P_p = E_{out} / t_p$.

In this study the Q-switched laser is pumped by a cw diode laser. In the repetitively Q-switched laser system the inversion population rises from a final value n_f after a Q-switch pulse is generated to an initial value n_i before a Q-switch pulse starts. For a four-level system the n and γ in Eqs. (3-1) and (3-2) are set to n_2 and 1, respectively. The buildup of the inversion under the influence of a continuous pumping rate and spontaneous decay rate is obtained as a function of time by

$$n(t)=n_\infty-(n_\infty-n_f)\exp\left(\frac{-t}{\tau_f}\right) \quad , \tag{3-15}$$

where n_∞ is the asymptotic value of the population inversion which is approached as t becomes large compared to the spontaneous decay time τ_f. The value n_∞ which depends on the absorbed pump power, is reached only at repetition rates small compared to $1/\tau_f$. For repetitive Q-switching at a repetition rate f, the maximum time available for the inversion to build up between pulses is $1/f$. Therefore,

$$n_i = n_\infty - (n_\infty - n_f)\exp\left(\frac{-1}{\tau_f f}\right) \quad , \tag{3-16}$$

in order for the inversion to return to its original value after each Q-switch cycle.

During each cycle a total energy $(n_i - n_f)h\nu$ contributes to generate the coherent electromagnetic field. Of this a fraction, $T/(T+\delta)$, appears as laser output. Therefore, the peak power P_p of the Q-switched pulses is

$$P_p = \frac{-\ln R}{8} \frac{\pi w_o^2}{\tau_c \sigma} h\nu (r - 1 - \ln r) \tag{3-17}$$

were w_o^2 is the mode spot waist.

Laser Design

The QPM grating of 13.6-μm period in a 17-mm long, 1-mm wide, and 1-mm thick z-cut Nd:MgO:$LiNbO_3$ crystal was fabricated by using the high-temperature (160°C) electric-field poling technique [31]. The characterized coercive field of this crystal was around 1.4 kV/mm with extrapolation of domain reversal current [32]. The co-doping concentrations of Nd and MgO in the Nd:MgO:$LiNbO_3$ crystal are 0.4- and 5-mol. %, respectively. The 13.6-μm grating period for the Nd:MgO:PPLN crystal is designed to phase match the 1st order EO PM QPM process [10] for a 1.085-μm light at 40°C.

In this dissertation, the commonly micro-lithographic process was taken. On the +z surface of a Nd:MgO:$LiNbO_3$ wafer, spin coated photoresist AZ4620 was used to pre-define the grating pattern as shown in Fig. 3.1, and a square opening as contact window was also included for connecting the positive electrode from high-voltage power supply. There were two parallel stripes beside grating pattern to conduct voltage from contact window to grating pattern. Then the photo patterned Nd:MgO:$LiNbO_3$ was covered NiCr alloy thin film about 80-nm-thick as the metal electrode by a RF sputter evaporator. The poling configurations are shown in Fig. 3.2. Immersed the Nd:MgO:$LiNbO_3$ sample into acetone organic solvent to perform lift-off process, and the designed grating electrode for poling the Nd:MgO:PPLN was done. For the poling process the wafer was sticked on a polished conductive plate with -z surface contacting the plate, and both of them were immersed into the silicon oil which was heated with a 5°C/min heating rate to 160°C (surface temperature of conductive plate) to decrease coercive field and to slow down the domain sideway expansion during domain inversion process. Oil temperature was controlled via digital thermal hot plate with external thermal-couple carefully because the coercive field in such high temperature is quite sensitive according to fabrication experiences. The poling waveform was generated via a computer controlled NI-DAQ adapter interface, and was amplified 2000 times by a high voltage power supply (Trek Inc. model 20/20C) to output poling voltage. Positive electrode was connect to the contact windows on +z surface of sample, and negative electrode was connect to conductive plate.

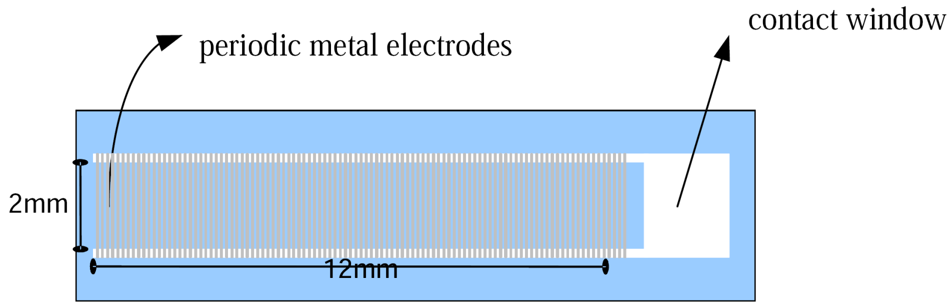


Fig. 3.1 There are the grating pattern and a square opening as contact window included for connecting the positive electrode from high-voltage power supply. There are also two parallel stripes beside grating pattern to conduct voltage from contact window to grating pattern.

The current response of MgO-doped advanced crystals including Nd:MgO:$LiNbO_3$ is different from $LiNbO_3$ crystal that was observed in previous research articles [32]. The mechanism of this phenomenon is still not clearly explained, however, it did cause difficulty for transferring $LiNbO_3$ poling philosophy to MgO-doped advanced $LiNbO_3$ crystals. That is why the poling processes have to be modified. The basic domain kinetics including domain nucleation at the electrode edge, domain tip propagation toward the opposite face of the crystal, termination of the tip at the opposite side of the crystal, rapid coalescence under the electrodes, sideway propagation of he domain wall, and the stabilization of the new domains are happened during the electric field poling. Before initiating poling the needed electric charge output has to be calculated from $Q=2P_sA$, where Q is the amount of electric charge, P_s is the spontaneous polarization density which is a material property, and A is the area to be poled. For this crystal, P_s is about 80 μC/cm^2. In this work the poling field is set to 2.1 kV/mm at 160°C. Fig. 3.3 shows the electric field and current versus time respectively.

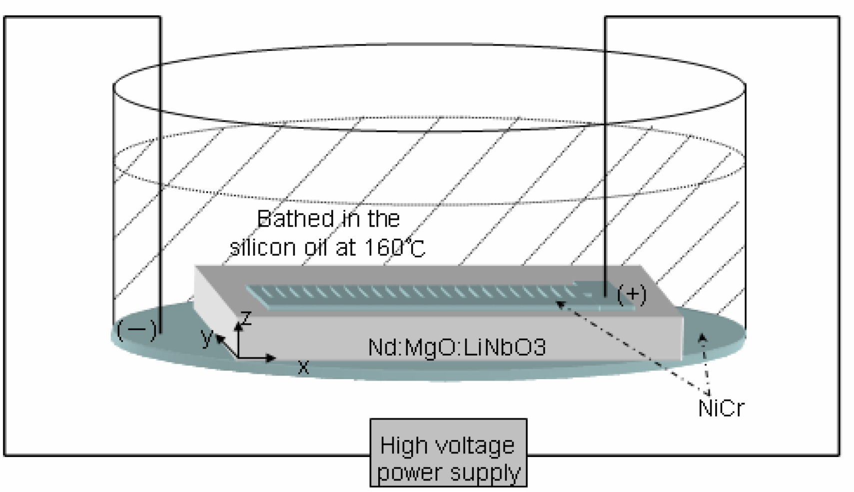


Fig. 3.2 The poling configurations are shown. For the poling process the wafer is sticked on a polished conductive plate with -z surface contacting the plate, and both of them are immersed into the silicon oil which was heated with a 5°C/min heating rate to 160°C (surface temperature of conductive plate) to decrease coercive field and to slow down the domain sideway expansion during domain inversion process.

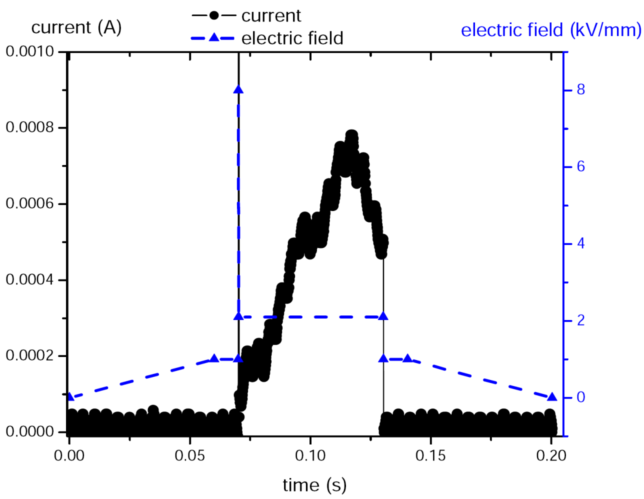


Fig. 3.3 The electric field and corresponding current versus time respectively.

Fig. 3.4 (a) and (b) show two typical cross-sectional photographs of the HF-etched *y* surface of the fabricated Nd:MgO:PPLN crystal taken near the former +z surface and at the depth of around 600 μm from the former +z surface, respectively. The photographs indicate that a uniform periodicity is maintained from the crystal surface only to a ~600 μm depth. The reason for the occurrence of this earlier domain termination or merging can be attributable to the peculiar resistance reduction problem found in electric-field domain-polarity switching of a $MgO:LiNbO_3$ crystal [33]. Nevertheless, a Nd:MgO:PPLN crystal with a ~600-μm clear aperture is already sufficient for current applications (see below). The average domain duty cycle estimated from the periodically poled region is ~30%/70%. The first 5-mm length of this Nd:MgO:PPLN crystal by design serves as the diode end-pump region where the diode pump power is predominantly deposited due to the exponential absorption law. This mitigates the thermo-optic effects in the downstream 12-mm long EO PM QPM device whose performance is sensitive to changes in the crystal birefringence and refractive indices. The two *y* faces of the 12-mm long EO Nd:MgO:PPLN device were sputtered with NiCr metal alloy thin film layers as side electrodes for building up electric field along *y* direction. The device functioned as a Pockels' cell [5] and served as the laser's Q-switch in this work.

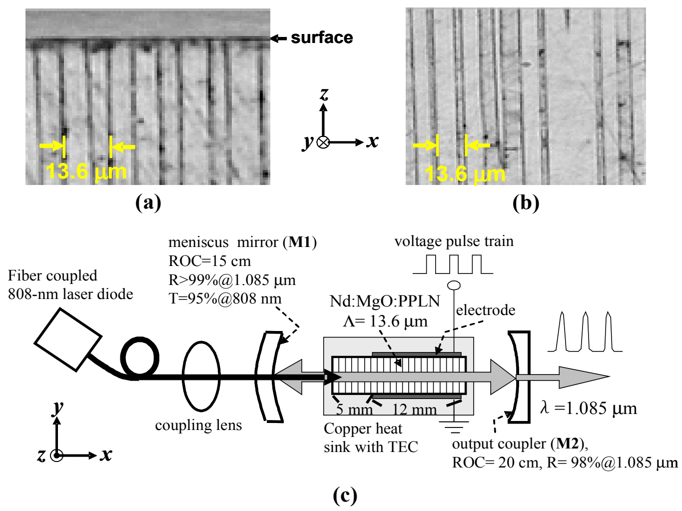
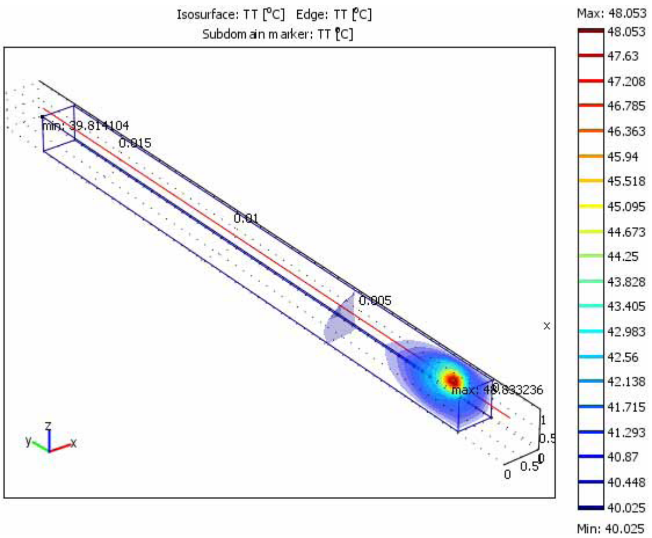


Fig. 3.4 (a) and (b) show two typical cross-sectional photographs of the HF-etched *y* surface of the fabricated Nd:MgO:PPLN crystal taken near the former +z surface and at the depth of around 600 μm from the former +z surface, respectively. Scheme of a compact diode end-pumped, internally Q-switched Nd^{3+} laser based on the fabricated monolithic Nd:MgO:PPLN device is shown in (c).

Then a compact diode end-pumped, internally Q-switched Nd^{3+} laser based on the fabricated monolithic Nd:MgO:PPLN device was constructed, as schematically shown in Fig. 3.4(c). The pump laser was an 808-nm fiber-coupled diode laser with an emission polarization ratio of 2.5:1. The 808-nm wavelength is coincident with a σ-polarized wave absorption band of a Nd:MgO:$LiNbO_3$ crystal [34]. A coupling lens set was used to collimate and focus (with an effective focal length ~20 mm) the pump beam into the Nd:MgO:PPLN crystal near the front surface. The pump waist radius was ~190 μm. The lower halves of both crystal end faces (*x* surfaces) have been blocked to ensure lasing through the upper 600-μm periodically poled region. Both of the crystal end faces have the anti-reflection (AR) coatings for dual wavelengths at 808 nm and 1085 nm. The Nd:MgO:PPLN crystal was first mounted in a copper heat sink whose temperature was stabilized by a thermal-electric cooler (TEC) unit, then placed in a laser resonator. The resonator was composed of a 15-cm radius of-curvature (ROC) meniscus (zero power) dielectric mirror (M1) having >99% reflectance at 1.085 μm and ~95% transmittance at 808 nm and a 20-cm ROC plano-concave mirror (M2) having 98% reflectance at 1.085 μm as the laser output coupler. The cavity length of the constructed Nd:MgO:PPLN laser was as compact as 3.4 cm. The Nd:MgO:PPLN laser radiates at π-polarized 1.085 μm through the $^4F_{3/2} \rightarrow {}^4I_{11/2}$ transition. The measured cw threshold absorbed pump power and slope efficiency for this laser were 318 mW and ~11%, respectively. The laser output mode was in a slightly elongated shape with an eccentricity of ~0.3. The characterized M^2 values approached 1 for both the vertical and horizontal beam divergence directions. These results can be attributable to the geometry of the intracavity Nd:MgO:PPLN crystal (~600 μm in clear aperture and 17 mm in length as aforementioned) that poses the emission of an elongated TEM00 beam. The ratio of the peak emission cross sections of the π-polarized to the σ-polarized lines of the $^4F_{3/2} \rightarrow {}^4I_{11/2}$ transition in a Nd:MgO:$LiNbO_3$ crystal is ~3.5 [34], which is quite close to that (~3.8) of the same transition in a Nd:YVO_4 crystal (note, however, the cross-section values of Nd:YVO_4 are more than one order of magnitude larger than those of $Nd:MgO:LiNbO_3$ for that transition). This large gain ratio between the two polarizations facilitates the efficiency of operation of an EO PPLN device as a laser Q-switch [5,7,10].

Based on the pumping scheme and the TEC temperature controlling system specifications, the thermo-optic effect was investigated on the fabricated monolithic Nd:MgO:PPLN device of performing an EO internal Q-switch. Fig. 3.5 shows the isosurfaces of the temperature inside the crystal calculated by a finite-element method software, Comsol Multiphysics , under an absorbed pump power of 0.61 W and a TEC stabilizing temperature of 40°C. It was estimated that ~31% of the incident 808-nm pump power is converted to heat due mainly to the Stokes shift (η_S~ 0.74) and pump quantum efficiency (η_Q~ 0.93) of a $Nd:MgO:LiNbO_3$ laser [35]. All the absorbed power turned into the sum of deposited thermal power and laser power was considered. Therefore, the calculation result in Fig. 3.5 is overestimated since the fluorescence loss was also considered as the thermal load [36][37]. It can be clearly seen from Fig. 3.5 that the temperature change due to this thermal deposition in the crystal is mainly located at the very first 5 mm of the crystal. The red line indicates the optical axis of the laser system. The minimum temperature is ~39.8°C located at the far end of the crystal and the maximum temperature is ~48.8°C located at about 1 mm from the crystal front surface, corresponding to the focus of the pump beam. Fig. 3.6 shows the temperature distribution (solid curve) along the optical axis in the Nd:MgO:PPLN crystal. The corresponding distributions of the refractive index change of the crystal for 1.085-μm σ-polarized and π-polarized waves (dotted and dashed curves, respectively) are also plotted in Fig. 3.6 using the Sellmeier equations of a $MgO:LiNbO_3$ [38].



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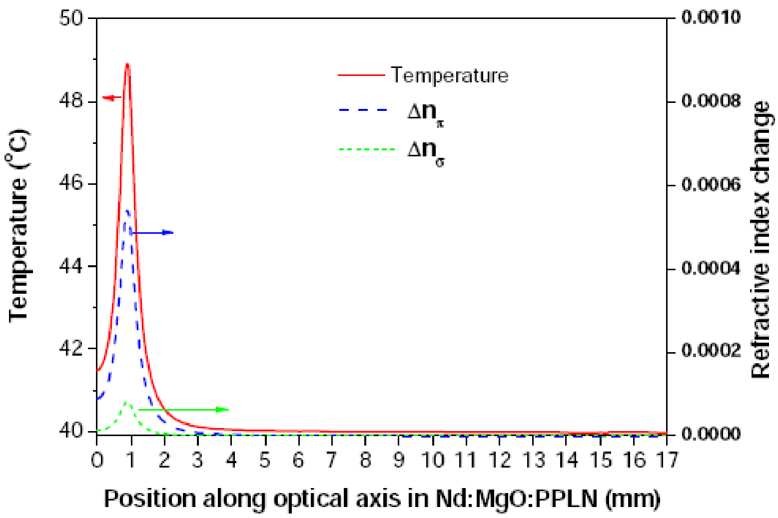
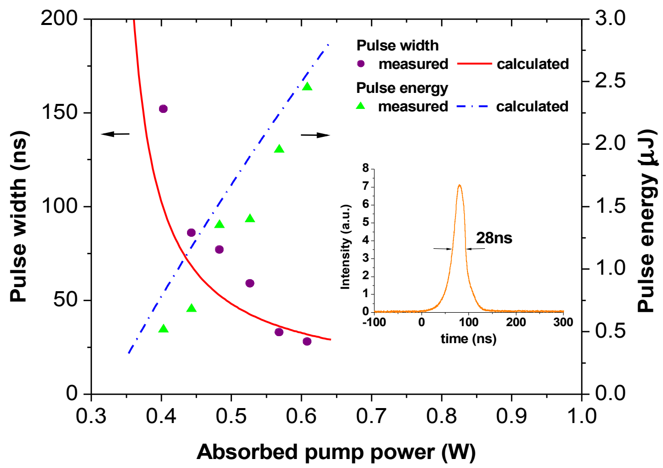


Fig. 3.6 It shows the temperature distribution (solid curve) along the optical axis in the Nd:MgO:PPLN crystal. The corresponding distributions of the refractive index change of the crystal for 1.085-μm σ-polarized and π-polarized waves (dotted and dashed curves, respectively) are also plotted in this figure using the Sellmeier equations of a MgO:$LiNbO_3$.

Laser Performance

To operate the laser in Q-switching mode, the EO Nd:MgO:PPLN section was first driven with a 5-kHz voltage pulse train with 260 V in amplitude and 300 ns in pulse width. This 260-V driving voltage corresponds to a normalized value of 0.31 V×d (μm)/L (cm), where d (=1000 μm for the present device) is the electrode separation and L (=1.2 cm for the present device) is the electrode length. The demand of a higher driving voltage than that for an undoped PPLN (<0.18 V×d (μm)/L (cm)) [5] can be attributable to the lower domain poling quality obtained with our current technical capability for implementing a uniform periodicity and 50%-duty cycle Nd:MgO:PPLN crystal. At 0.61-W absorbed pump power, we obtained laser pulses of pulse energy >2.45 μJ and a pulse width <30 ns, corresponding to a laser peak power of ~88 W. Fig. 3.7 shows the variation of the measured pulse width (solid circles) and pulse energy (solid triangles) with the absorbed pump power. The pulse width decreased from 152 to 28 ns and the pulse energy increased from 0.52 to 2.45 μJ as the absorbed pump power increased from 0.4 to 0.61 W. The inset shows the measured Q-switched laser pulse. By using the material properties of a Nd:MgO:$LiNbO_3$ crystal [34] (absorption coefficient α = 0.002 cm^{-1} , upper level lifetime τ_f ~ 100 μs, and emission cross section $\sigma_s \sim 2\times10^{-19}$ cm^2 at λ_L =1.085 μm), the aforementioned cavity geometric and coating parameters, and the measured pump threshold and slope efficiency (η_{sl}) from the cw laser output, the theoretical fittings (solid and dash-dotted curves, respectively) were calculated for the two measurements plotted in Fig. 3.7 according to the model to predict the output performance of an EO PPLN Q-switched Nd:YVO_4 laser [10]. The peak power and pulse width as a function of the Q-switch repetition rate were also measured with the EO Nd:MgO:PPLN laser system operated at an absorbed pump power of 0.57 W; indicated in Fig. 3.8 by the solid triangles and circles, respectively. The pulse width increased from 33 to 160 ns and the peak power decreased from 59.3 to 1.6 W as the Q-switch repetition rate increased from 5 to 35 kHz. In this figure, the solid and dash-dotted curves represent the theoretical fittings of the two measurements, obtained using the aforementioned system parameters and calculation model. It can be seen from Figs. 3.7 and 3.8 that the experimental data agrees reasonably well with the theoretical predictions. This agreement implies that the 2% output coupling used in this work is close to the optimum value for the present laser system in the applied diode pump power range [39]. To further enhance the output performance (e.g., the pulse width reduction and pulse energy extraction) of this internal-Q-switched laser, the increase of the output coupler transmittance is necessary. To optimize the Q-switched laser system with a higher output coupling value, the pump threshold of the system has to be lowered accordingly [39]. This can be done by reducing the dissipative loss of the resonator and/or the pump beam size (it is ~380 μm with current setup) [34]. To reduce the pump beam size in the Nd:MgO:PPLN crystal, a more compact design of the laser system might be helpful. For example, one can directly apply a high-reflectivity mirror coating on the input end of the Nd:MgO:PPLN crystal to save the mirror M1 (refer to Fig. 3.4(c)).

A few-watt (peak power) non-phase-matched second harmonic generation (542 nm) from this laser system was also observed. No photorefractive effects were observed in the experiment. This appreciable amount of generation originates in the high intracavity fundamental peak power (~4.4 kW). This observation suggests the laser system has the potential to develop as an interesting internal-Q-switched self-frequency-conversion laser if an additional QPM wavelength-conversion section can be monolithically integrated into the current Nd:MgO:PPLN crystal.



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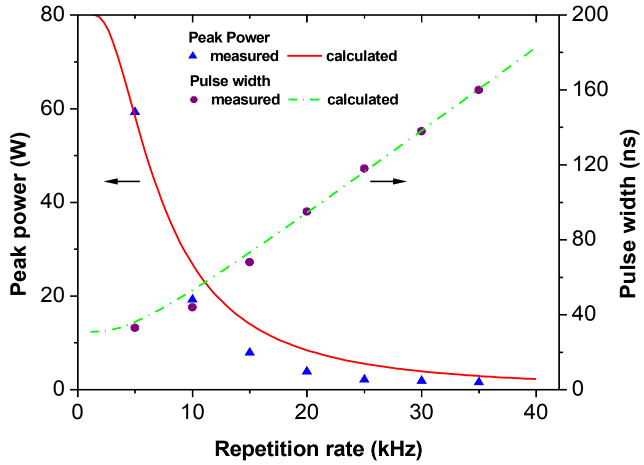

Fig. 3.8 The peak power and pulse width as a function of the Q-switch repetition rate were measured with the EO Nd:MgO:PPLN laser system operated at an absorbed pump power of 0.57 W; indicated by the solid triangles and circles, respectively. The pulse width increased from 33 to 160 ns and the peak power decreased from 59.3 to 1.6 W as the Q-switch repetition rate increased from 5 to 35 kHz. In this figure, the solid and dash-dotted curves represent the theoretical fittings of the two measurements, obtained using the aforementioned system parameters and calculation model.

Summary

A laser-diode-pumped, actively internal-Q-switched laser system has been successfully fabricated and demonstrated using a 17-mm long, 13.6-μm period monolithic Nd:MgO:PPLN crystal. 28-ns, 2.45-μJ Q-switched laser pulses, corresponding to a laser peak power of ~88 W were obtained from this miniature 1.085-μm laser system with 2% output coupling when driven by a 5-kHz 260-V voltage pulse train and pumped at 0.61-W absorbed diode power. The characterized output performance of this laser system agreed reasonably well with the theoretical calculations.

Chapter 4 EO Aperiodically Poled Lithium Niobate Devices

Introduction

Optical wavelength filters are key elements in many optical systems for the processing of the spectral content of optical signals. Narrowband optical filters are particularly useful in the fields of optical wavelength-division multiplexing (WDM) communications. Among the optical filter technologies, multilayer optical thin-film coating [40] could be the most popular one. However, thin-film filters operate passively with fixed spectral characteristics. Obviously, an active and tunable optical filter would be useful in applications requiring spectral adjustment. Šolc filters [41] are a kind of narrowband birefringence filter whose spectral transmission characteristics can tuned by changing the optical path difference or by varying the crystal angle to change the phase-retardation-matching condition in the filter. A conventional Šolc filter built in a single-domain birefringence crystal has to adopt a periodic electrode configuration to modulate the sign of the relevant EO coefficient so as to convert the polarization modes of the waves in it [42][43]. Since the sign of the EO coefficient in a periodically pole crystal is periodically changed due to the periodic reversal of the ferroelectric domain, a Šolc filter built into such a crystal can have a simpler electrode configuration to prevent short circuit possibilities and to permit the use of the full device length for further narrowing the spectral bandwidth. It is the demonstration of Šolc filters in PPLN crystals that was recently reported with emphasis on single-wavelength spectral tuning [4] and power modulation [5].

Unlike the uniform grating structure commonly adopted by a PPLN crystal, an aperiodic grating can be implemented into a lithium niobate crystal to provide more than one grating vector from the spatially modulated domain lattice. The multiple grating vectors can be designed to simultaneously compensate for several wave-vector mismatches that arise from multiple wave-energy coupling processes in an optical medium. Such a scheme has been adopted by the so-called QPM nonlinear wavelength conversion. For example, the concept of using an aperiodic grating structure has been applied to the design of a broadband QPM second-harmonic generator with the assistance of the simulated-annealing (SA) algorithm [44][45]. More recently, aperiodically poled lithium niobate (APLN) crystals were also used for demonstrating equal-gain multi-wavelength second-harmonic generations [14]. This chapter introduces the successful implementation of the SA algorithm to the design of multi-wavelength Šolc filters based on APLN crystals and demonstrate the electro-optically active narrowband multi-wavelength APLN filters for use in the telecom C band (~1530-1570nm).

Device design and simulation

When modeling the EO APLN filter, the filter was assumed that it consists of a sequence of N crystal domain blocks, and each of these blocks is with a thickness of Δx and a domain polarity of either +1 or −1 denoting the +z or −z crystal orientation of the block, respectively. Consider the ordinary and extraordinary polarization modes that propagate along the APLN crystallographic x axis with the wave numbers β_o and β_e, respectively. When an external electric field E_y is presented along the APLN crystallographic y axis, the perturbed part of the dielectric tensor of the crystal becomes aperiodically modulated along the x direction due to the sign reversal of the relevant Pockels' coefficient γ_{51} upon the reversal of the crystal domain in z. In the limit of weak perturbation, the power exchange between the two polarization modes in such a dielectric modulated structure can be described by a pair of co-directional coupled-mode equations which were introduced as Eq. (2-27).

To perform a proof-of-principle experiment, An EO APLN wavelength filter was designed to transmit 8 International Telecommunication Union (ITU) wavelengths, λ_1 = 1541.34 nm, λ_2 = 1543.73 nm, λ_3 = 1546.11 nm, λ_4 = 1548.51 nm, λ_5 = 1550.91 nm, λ_6 = 1553.32 nm, λ_7 = 1555.74 nm, and λ_8 = 1558.17 nm, at 37.5°C with a 300-GHz frequency separation between adjacent channels. In the computer codes, The parameters were set to $T_0(\lambda_\alpha)$= 1 and $w(\lambda_\alpha)$= 1 for α = 1, 2, 3, …,8 to maximize the transmittances at all the 8 designated wavelengths with equal weighting. Although a short domain thickness Δx is preferred for optimizing the device performance, Δx = 5 μm is chosen to meet the ability in $LiNbO_3$ crystal poling at that time. The total number of the domain blocks was set to be 10000 so that the length of the EO APLN filter was 5-cm. In fact, as small a Δx value as possible is always preferred in providing good amount of reciprocal vectors for stimulated annealing an optimum structure for prescribed multiple sets of couplings. Fig. 4.1 shows the algorithm flow chart that is planned to employ in programming and calculating an optimum aperiodic domain structure for an EO APLN for filtering prescribed multiple wavelengths in telecom C band based on the SA method, where the objective function *Ob* is used to guide the algorithm to both maximize and equalize the efficiencies of the prescribed *M* coupling processes of wavelengths λ_α with a = 1, 2, 3,…, *M*, given by

$$Ob=\{\sum_{\alpha=1}^{M}|\eta_0(\lambda_\alpha)-\eta_{cal}(\lambda_\alpha)|w(\lambda_\alpha)\}+\beta\{[\eta_{cal}(\lambda_\alpha)]_{max}-[\eta_{cal}(\lambda_\alpha)]_{min}\} \quad , \tag{4-1}$$

where $\eta_0(\lambda_\alpha)$, $\eta_{cal}(\lambda_\alpha)$, and $w(\lambda_\alpha)$ are the prescribed efficiency, the calculated efficiency, and the weighting factor of the wave λ_α, respectively, β is an adjustable parameter for efficiency equalization, and the footnotes *max* and *min* mean the maximum and minimum value among calculated efficiencies.

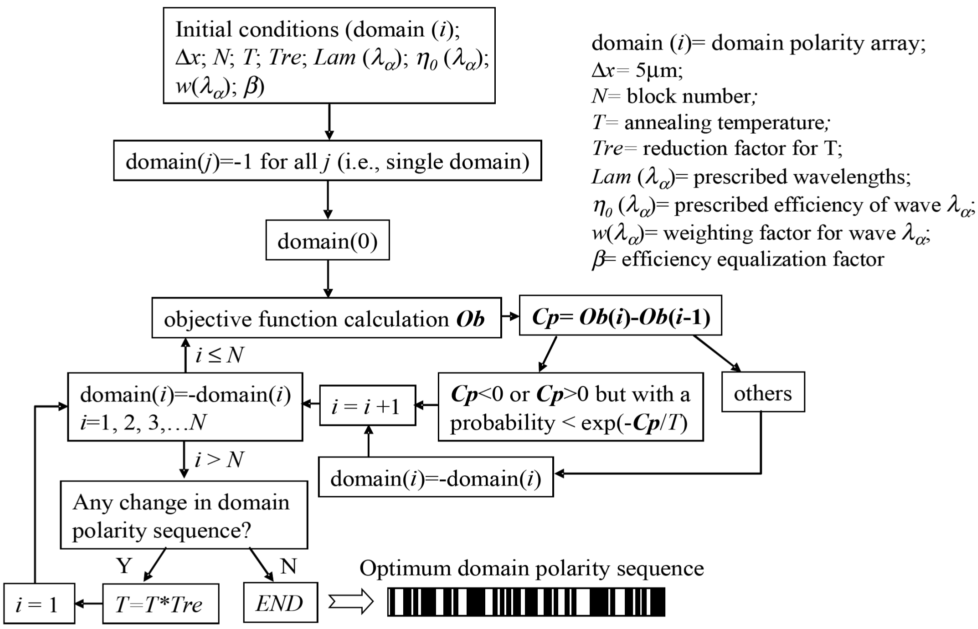

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Fig. 4.2(a) shows the calculated transmittance of the 5-cm long APLN filter (green curve) as a function of the applied electric field, from which we determine an optimum driving field of 277 V/mm at the peak transmittance. The green curve in Fig. 4.2(b) shows the transmittance of the filter as a function of the input wavelength at the optimal driving field, indicating nearly 100% transmittance for all 8 channels in our design. According to the green curve in Fig. 4.2(b), the noise floor for polarization cross-talk is approximately 25 dB lowered from the peak transmittance. For slowly varying field envelopes in the filter, the output field amplitudes are proportional to the Fourier transform of $s(x)$ evaluated at the spatial frequency $\Delta\beta = \beta_o - \beta_e$, according to Eqs. (2-27) and (2-30). Therefore the applied field E_y has a negligible effect on the transmission wavelengths of the filter. However, the coupling coefficient κ and thus the transmittances are dependent on E_y. As shown by the black dashed curve in Fig. 4.2(b), the transmission spectrum of the filter driven by an intermediate field of 100 V/mm shows the same peak transmission wavelengths but has lower peak transmittances.

In addition, our calculation predicts a signal transmission spectral width of ~0.45 nm for each of the 8 channels at the optimal driving field in the EO APLN filter. We found that the transmission width of a 5-cm long PPLN Šolc filter is also about 0.45 nm, despite that a PPLN filter only transmits a single wavelength. If one would cascade 8 equal-length segments of PPLN, each having a length of 5/8 cm and a suitable grating period, to form a 5-cm long Šolc-type filter for transmitting the same 8 ITU wavelengths, this PPLN filter would not only have a much higher driving field but also become impractical due to the crosstalk among the broadened channel peaks. The blue curve in Fig. 4.2(a) shows the calculated transmittance of a 5/8-cm long single-segment PPLN filter phase-matched at 1548.51 nm as a function of the applied electric field. It is evident from the figure that the driving electric field at peak transmittance for the PPLN filter is increased by 2.4 times from that of the APLN filter. For comparison, we also plot in Fig. 4.2(b) the transmission spectra of the single-segment PPLN (blue dash-dotted curve) and the 8-segment PPLN (red dotted curve) at the peak-transmission field of 675 V/mm (refer to the blue curve in Fig. 4.2(a)). It can be seen from the figure that the spectral width of the single-channel PPLN filter is 8.4 times larger than that of the APLN filter. Furthermore, the severe crosstalk and the decreased transmittance render the cascaded PPLN filter useless. The comparison in Fig. 4.2 indeed unambiguously illustrates the superior performance of an EO APLN filter over a cascaded EO PPLN filter.

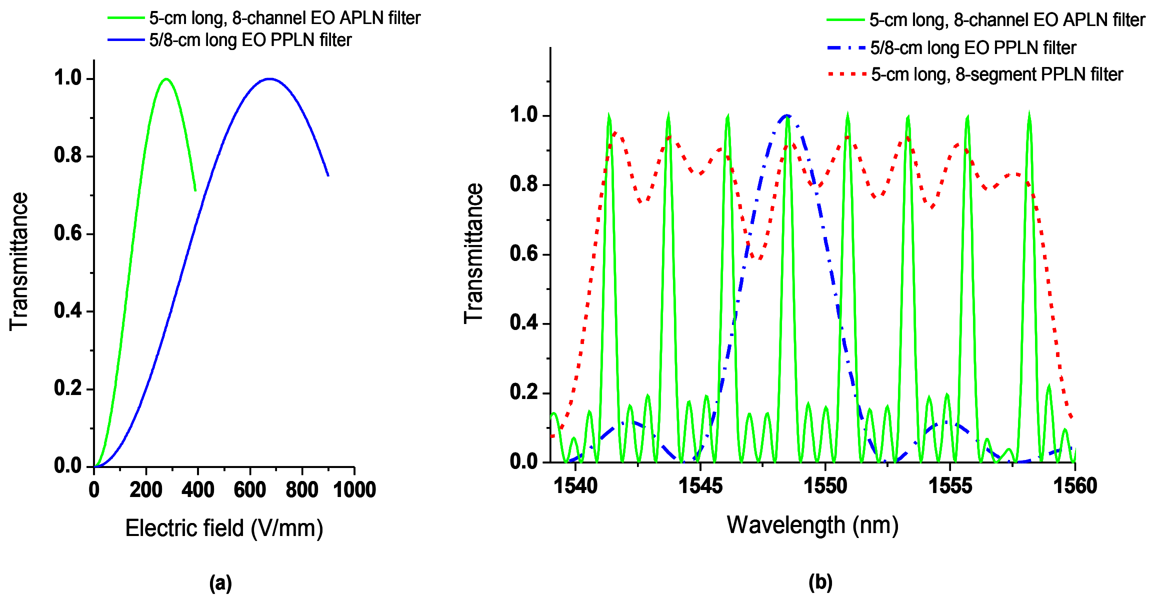


Fig. 4.2 (a) It shows the calculated transmittance of the 5-cm long APLN filter (green curve) as a function of the applied electric field, from which we determine an optimum driving field of 277 V/mm at the peak transmittance. The green curve in (b) shows the transmittance of the filter as a function of the input wavelength at the optimal driving field, indicating nearly 100% transmittance for all 8 channels in our design. According to the green curve in (b), the noise floor for polarization cross-talk is approximately 25 dB lowered from the peak transmittance.

Experimental demonstration and discussion

The APLN crystal was fabricated by using the conventional electric-field poling technique [18]. The fabricated APLN multi-wavelength filter has a dimension of 5 cm in length, 1 mm in width, and 0.5 mm in thickness. The end faces of the APLN crystal were optically polished but were without any optical coating. The two *y* faces of the APLN crystal were sputtered with NiCr alloy thin film to form side electrodes. The performance of the EO APLN filter was then characterized by using a single-mode, fiber-coupled amplified-spontaneous-emission (ASE) light source in the telecom C band. The fiber from ASE was connected to a in-line polarizer plus a polarization maintained fiber to ensure the light remain linear polarized in the designated direction. The output beam of the ASE source was collimated and size-reduced to a radius of ~110 μm in the crystal to prevent refractive losses. The APLN crystal was mounted in a temperature controlled oven with an analyzer at the output end of the crystal. The transmission axes of the polarizer and the analyzer were aligned and calibrated along the *y* and *z* axes of the APLN by power meter measurement, respectively. Fig. 4.3 shows the calculated (solid line) and measured (open circles) transmission spectra of the 5-cm long EO APLN filter at 37.5ºC. The inset shows a microscopic image of an HF-etched *z* surface of the fabricated APLN crystal, which indicates the aperiodicity of the domain structure synthesized from the 5-μm long domain blocks. By inspecting the resulting sequence function $s(x)$ or the fabricated APLN crystal, it is found that the aperiodic domain structure was composed of 3 different domain thicknesses Δx= 5 μm, 2Δx= 10 μm, and 3Δx = 15 μm. Most of the constituent domains have the thickness of 10 μm, which is consistent with the QPM coherence length of 10.62 μm for an EO PPLN device working in the telecom C band [46]. The experimental data in Fig. 4.3 indicate >90% peak transmittance measured for all the 8 wavelength channels. Although the location and width of each channel are in good agreement with the theoretical predictions, the side lobes in the measured data appear to be more apparent than those in the calculated spectrum. To understand the slight discrepancy between the experimental and calculated results, a set of computer simulations similar to that in ref. [14] were first conducted to account for the domain-width variations incurred during the electric-field poling process. The simulations indicate that the EO APLN filter is insensitive to this type of fabrication errors, as also concluded in ref. [14] for APLN wavelength converters. The effect of the off-axis incidence of a Gaussian laser beam was investigated as well. The off-axis incident angle changes the spatial harmonic in $s(x)$ and shifts the transmission spectrum. For the laser beam to stay within the 1-mm crystal width over a length of 5 cm, the largest possible incident angle for a 110 μm-waist laser beam is about 15 mrad, as illustrated by the upper-left inset in Fig. 4.4(a). This worse-case output spectrum of the EO APLN filter was calculated and shown in Fig. 4.4(a) (green curve). The off-axis incidence of the laser beam indeed causes a slight spectral shift of ~+0.17 nm, but does not give a noticeable change to the peak transmittances. A similar conclusion can be drawn from the estimate of the angular acceptance bandwidth of a QPM nonlinear optical material due to the beam walk-off effect. Following the similar derivation in [47], the angular acceptance bandwidth is obtain for 1%

reduction from peak transmittance in an EO PPLN filter, given by

$$\delta\varphi_{1\%} = 2\sqrt{\frac{0.6296}{\pi}\frac{n_e}{n_o}\frac{l_c}{L}\cos\varphi} \quad , \tag{4-2}$$

where φ is the laser incident angle relative to the normal to the crystal surface, L is the crystal length, and l_c is the coherence length of the EO QPM process. For this design $\varphi = 0$, $L = 5$ cm, and l_c ~10.62 μm for telecom C-band wavelengths, the angular acceptance bandwidth 1% $\delta\varphi$ in such an EO PPLN filter is approximately 15 mrad. Therefore the divergence and the off-axis incidence of the input laser beam can not be the only source causing the discrepancy. In our investigation, the discrepancy can be explained by a slight temperature gradient across the longitudinal direction of the 5-cm long APLN crystal in our homemade crystal oven. The phase matching wavelength similar to nonlinear process is altered via changing crystal temperature for a given grating period and can be expressed as

$$\frac{d\lambda}{dT} = \Lambda\left(\frac{d\,n_o(\lambda, T)}{dT} - \frac{dn_e(\lambda, T)}{dT}\right) \quad , \tag{4-3}$$

where λ is the phase matching wavelength in nm, Λ is the grating period, and T is the temperature in ºC. The refractive indices are functions of temperature according to the Sellmeier's equation,

$$n^2 = A_1 + \frac{A_2 + B_1 F}{\lambda - (A_3 + B_2 F)^2} + B_3 F - A_4 \lambda^2 \quad , \tag{4-4}$$

where $F = (T\text{-}24.5)/(T\text{+}570.5)$, for n_o, $A_1 = 4.9048$, $A_2 = 1.1775\times10^5$, $A_3 = 2.1802\times10^2$, $A_4 = 2.7153\times10^8$, $B_1 = 2.2314\times10^2$ $B_2 = -2.9671\times10^5$, $B_3 = 2.2971\times10^7$, and for n_e, $A_1 = 4.582$, $A_2 = 9.921\times10^4$, $A_3 = 2.109\times10^2$, $A_4 = 2.194\times10^8$, $B_1 = 5.2716\times10^2$, $B_2 = -4.9143\times10^5$, $B_3 = 2.2971\times10^7$.

The red curve in Fig. 4.4(a) shows the simulation result for the 5-cm long EO APLN filter with a temperature gradient of −0.1ºC/cm decreased from the center to both ends of the crystal. With that amount of temperature gradient, the red-curve spectrum in Fig. 4.4(a) indeed unveils the side lobes and reduced transmittance similar to those in the measured spectrum. To verify this temperature effect, we further fabricated another 1-cm long EO APLN filter designed for filtering four wavelengths in the telecom C band. Fig. 4.4(b) shows a comparison of the calculated and measured output spectra of this short EO APLN filter, indicating excellent agreement between the theory and experiment.

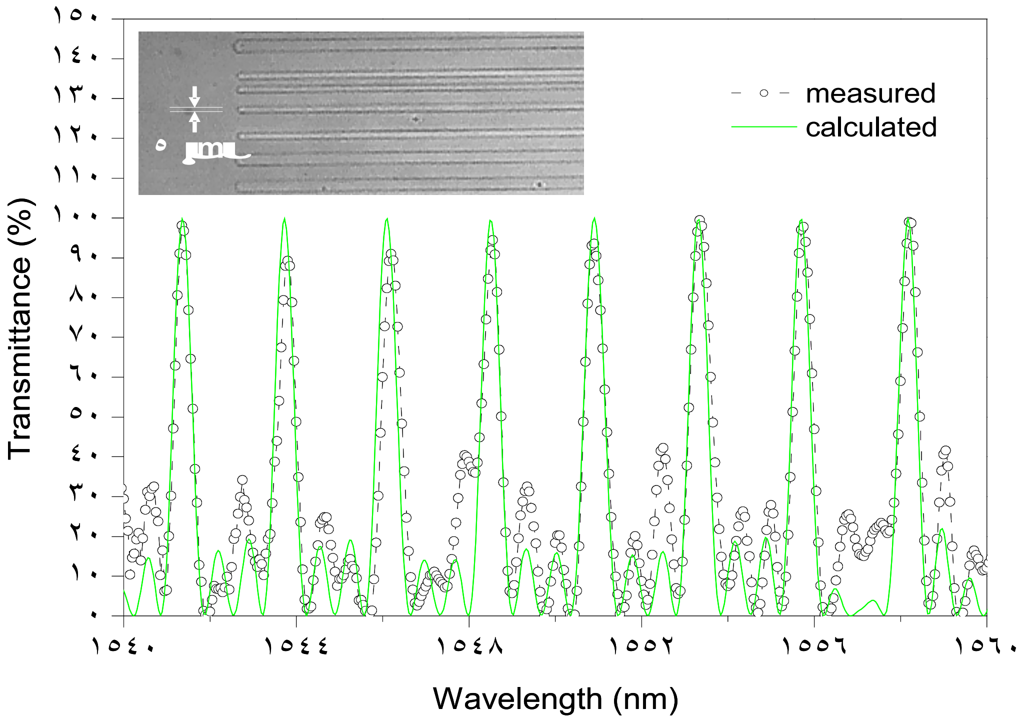


Fig. 4.3 It shows the calculated (solid line) and measured (open circles) transmission spectra of the 5-cm long EO APLN filter at 37.5ºC. The inset shows a microscopic image of an HF-etched *z* surface of the fabricated APLN crystal, which indicates the aperiodicity of the domain structure synthesized from the 5-μm long domain blocks. By inspecting the resulting sequence function $s(x)$ or the fabricated APLN crystal, it is found that the aperiodic domain structure was composed of 3 different domain thicknesses Δx= 5 μm, 2Δx= 10 μm, and 3Δx = 15 μm. Most of the constituent domains have the thickness of 10 μm, which is consistent with the QPM coherence length of 10.62 μm for an EO PPLN device working in the telecom C band. The experimental data indicate >90% peak transmittance measured for all the 8 wavelength channels.

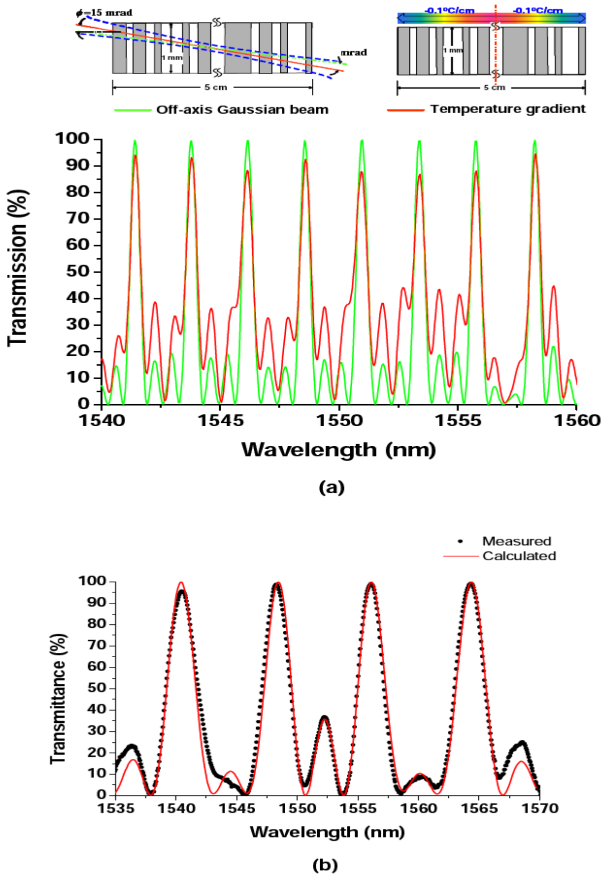


Fig. 4.4 The red curve in (a) shows the simulation result for the 5-cm long EO APLN filter with a temperature gradient of −0.1ºC/cm decreased from the center to both ends of the crystal. With that amount of temperature gradient, the red-curve spectrum in (a) indeed unveils the side lobes and reduced transmittance similar to those in the measured spectrum. To verify this temperature effect, we further fabricated another 1-cm long EO APLN filter designed for filtering four wavelengths in the telecom C band. (b) It shows a comparison of the calculated and measured output spectra of this short EO APLN filter, indicating excellent agreement between the theory and experiment.

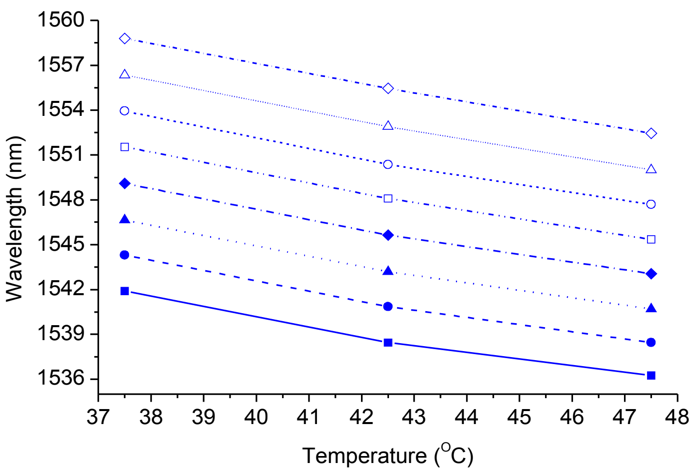


Fig. 4.5 It plots the measured spectral tuning as a function of temperature for the 8 transmitted wavelengths of the 5-cm long EO APLN filter. A tuning rate of ~0.65 nm/ºC can be derived from the plot.

Since the refractive indices of the crystal are temperature dependent, it is possible to tune the transmission spectrum of an EO APLN filter by varying the temperature. Fig. 4.5 plots the measured spectral tuning as a function of temperature for the 8 transmitted wavelengths of the 5-cm long EO APLN filter. A tuning rate of ~0.65 nm/ºC can be derived from the plot.

Conclusion

In conclusion, an active multi-wavelength optical filters by using EO APLN crystals have been successfully demonstrated. A derived coupled-mode theory was used to model the device behavior and to developed a computer code which is based on the simulated-annealing algorithm to optimize the design of such EO APLN filters. Fig. 4.2 clearly shows the superior performance of an EO APLN multi-wavelength filter over a cascaded EO PPLN filter. In experiment, there were simultaneously 8 ITU-standard wavelengths transmitting through a 5-cm long EO APLN filter with a transmittance of >90% and a bandwidth of ~0.45 nm for each wavelength channel. The transmission spectrum of the EO APLN filter can be conveniently tuned by temperature at a rate of ~0.65 nm/ºC in the telecom C band. The performance of this device is much more sensitive to the temperature gradient than to the domain-width variation in the crystal. Although the EO APLN filter is well demonstrated in the telecom C band, the application of such a filter is apparently not limited to optical communications. To operate at the TTL-level voltage, the EO APLN filter presented in this chapter could be implemented in, for example, a Ti-diffused $LiNbO_3$ waveguide [46]. The APLN wavelength filter can also be cascaded with an APLN or a PPLN wavelength converter to perform integrated optical signal filtering and mixing.

Chapter 5 APLN Q-switched and Frequency-doubled Laser

Introduction

In this dissertation the EO PPLN/APLN has been demonstrated for applications such as wavelength filter, and active laser Q-switch. For the wavelength filter, the length of EO PPLN crystal is longer, then the filtering bandwidth is narrower. For laser, EO PPLN can be placed in a laser cavity as a Q-switch just like a Pockels' Cell to introduce additional laser cavity losses via switching the laser polarization between high- and low-loss laser modes for polarization-dependent laser gain medium such as Nd:YVO_4, Nd:$LiNbO_3$ [2], and so on.

In the past few decades, PPLN has been proven as a practical QPM material for high-efficiency nonlinear frequency conversion due to its large effective nonlinear coefficient and free of walk-off problems in nonlinear wavelength conversion. QPM wavelength conversion efficiency is proportional to the input fundamental power and the interaction length. Hence such nonlinear optics devices were favorably placed in laser cavities to achieve much higher conversion efficiencies. However the conversion efficiencies could be still limited by the diffraction and clipping losses owing to the small aperture of PPLN bulk crystals (~0.5-1 mm) comparing with other conventional birefringence nonlinear crystals. Then the system alignment became more and more critical as intracavity PPLN length increases. Usually the aperture of PPLN is 500 μm to 1000 μm which is restricted by pretty high coercive field, over 21 kV/mm, of the congruent grown lithium niobate (CLN) single crystal. There is a practical method to achieve larger aperture QPM material that is to use MgO/ZnO doped lithium niobate whose coercive field is low down to 4.5 kV/mm in room temperature, but its cost is about 4 times higher than that of CLN one with the same wafer thickness.

Since PPLN can functions as laser Q-switch and QPM frequency converter, Chen and others have experimentally demonstrated an integrated QPM device can be versatile in function by ingenious integration of its electro-optic and wavelength-conversion advantages in a monolithic crystal substrate and can be readily implemented through nowadays mature micro-fabrication techniques [7]. But they just achieved both functions in a preliminary way that there was an EO PPLN as a laser Q-switch and a cascaded QPM PPLN as IOPO. In chapter 4 an APLN as a monolithic EO multiple wavelengths filters for WDM application have been demonstrated. Those works proved that with the same iterative computing algorithm for designing a QPM APLN multi-wavelength SHG device [14], an EO QPM APLN multi-wavelength filter can be realized as well. At the end we proposed that an EO APLN filter can be further cascaded with a QPM wavelengths converter, however the filters and converters were still settled in different sections of cascade APLN crystal. The precursor research inspired us that if one can integrate an EO Šolc filter like PM converter and a frequency doubler into an AOS with a single binary sequence, it can utilize limited crystal length more efficiently.

To realize this integrate multi-function device, the design principles in ref. [14] and chapter 4 with iterative computing algorithm [44,45] were adopted to design and to optimize needed ferroelectric domain sequence in lithium niobate crystal. The object was to design a 2-cm long APLN who is efficient to use the full chip length for active EO Q-switch and QPM SHG functions. As the voltage is applied, the laser cavity is in high-loss mode. As the voltage turns off, the intracavity frequency doubler is activated to generate 671 nm radiation. The main reason for designing the laser wavelength in 1342 nm rather than the popular 1064 nm is due to 1342 nm is in the telecom O band (1260-1360 nm).

Design

The details about EO PPLN and APLN modeling theories have been described in previous chapters. The development process of the design algorithm for optimizing two different QPM coupling processes in a single-grating-structure APLN devices is similar to that we have treated in chapter 4 except the objective function has to be modified as

$$Ob_{2p} = |\eta_0 - \eta_{EO} w_{EO} - \eta_{SHG} w_{SHG}| \tag{5-1}$$

where η_0, w_{EO}, and w_{SHG} are the prescribed efficiency, the weighting factor for EO PM conversion efficiency η_{EO} and the weighting factor for QPM SHG efficiency η_{SHG}, respectively. Fig. 5.1 shows the flow chart we propose to use in programming and calculating an optimum aperiodic domain structure for this APLN crystal.

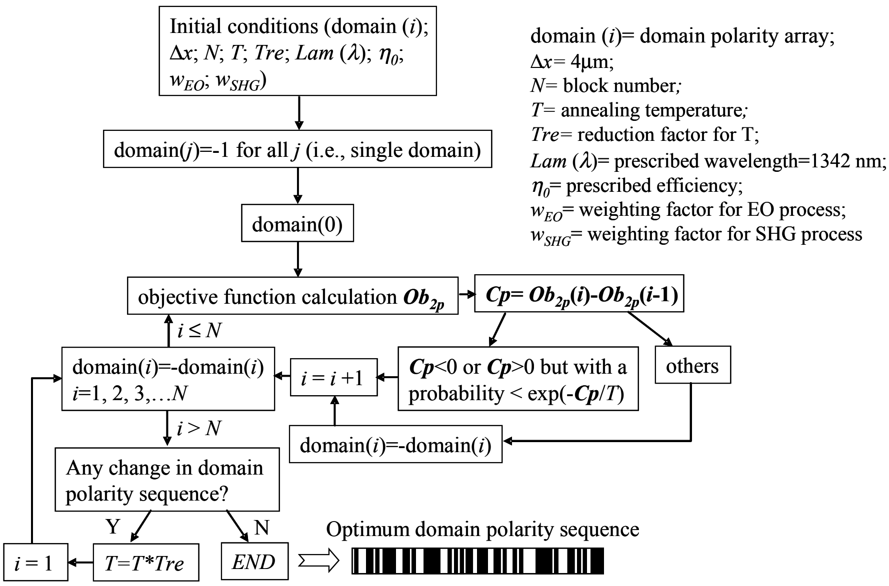


Fig. 5.1 The flow chart employed to calculate an optimum aperiodic domain structure for an APLN crystal for simultaneously being a laser Q-switch and a second-harmonic generator.

Base on these models we designed the object APLN and compared its performances with cascade multiple functions PPLN devices. For a 2-cm long EO PPLN it takes 195 V/mm to make the polarization of light wave at 1342 nm to turn 90°. For a 1-cm long EO PPLN it takes 390 V/mm to achieve 90°. Owing to that it is unnecessary to achieve 90° polarization mode change for Q-switching, we put the baseline electric field of APLN at 390 V/mm to make the SHG efficiency as high as possible and to keep the Q-switching voltage low. Taking into account the poling difficulty and the 1st order coherent length of QPM SHG process in PPLN for 1064 nm, 3.4 μm, and 1342 nm, 6.7 μm, the design minimum domain block width is 4 μm so the calculated domain widths are multiples of 4 μm. This also satisfies the demand for 1st order coherent length, 9 μm, of EO QPM PPLN at 1342 nm. Above conditions are calculated under 37°C.

In order to evaluate the grating vectors computed via iterative computing algorithm, not only the output of couple-mode equations were calculated but also the reciprocal vectors of APLN domains were analyzed. In Fig. 5.2 there are the calculated single-path SHG conversion efficiencies of the APLN and a 2-cm long bisection cascade PPLN which contents an EO PPLN and a QPM SHG PPLN. The SHG efficiency of the APLN is ~1.6 times higher than that of the PPLN. It indicates that the effective length of the SHG process in the APLN is longer than 1 cm. The reciprocal vectors analyzed results are shown in Fig. 5.3. Reciprocal vectors of the 2-cm long bisection PPLN are shown in Fig. 5.3 as well. From the altitudes of APLN it shows that there are only two dominant spatial frequencies which are mapping SHG and EO Q-switching respectively. The low frequency one ~0.055 μm^{-1} is mapping for EO QPM vector, and the high frequency one ~0.075 μm^{-1} is mapping for QPM SHG vector. These two frequencies are well matching two spatial frequencies from bisection PPLN. The altitudes, ~0.78, of APLN QPM SHG dominant frequency is higher than that, ~0.63, of PPLN QPM SHG frequency which means higher SHG efficiency. Simultaneously the altitudes of EO APLN and EO PPLN are close to 0.6 which means there are almost the same EO Q-switching electric fields for achieving the same losses.

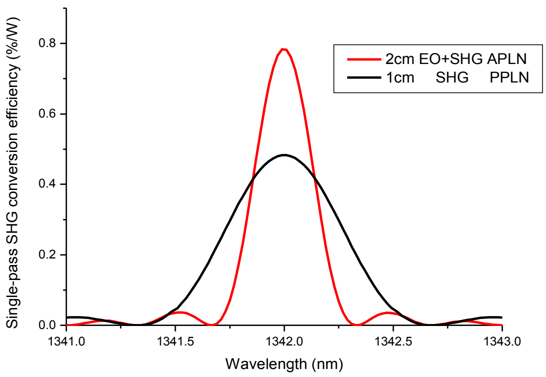


Fig 5.2 There are the calculated single-path SHG conversion efficiencies of the APLN and a 2-cm long bisection cascade PPLN which contents an EO PPLN and a QPM SHG PPLN. The SHG efficiency of the APLN is ~1.6 times higher than that of the PPLN.

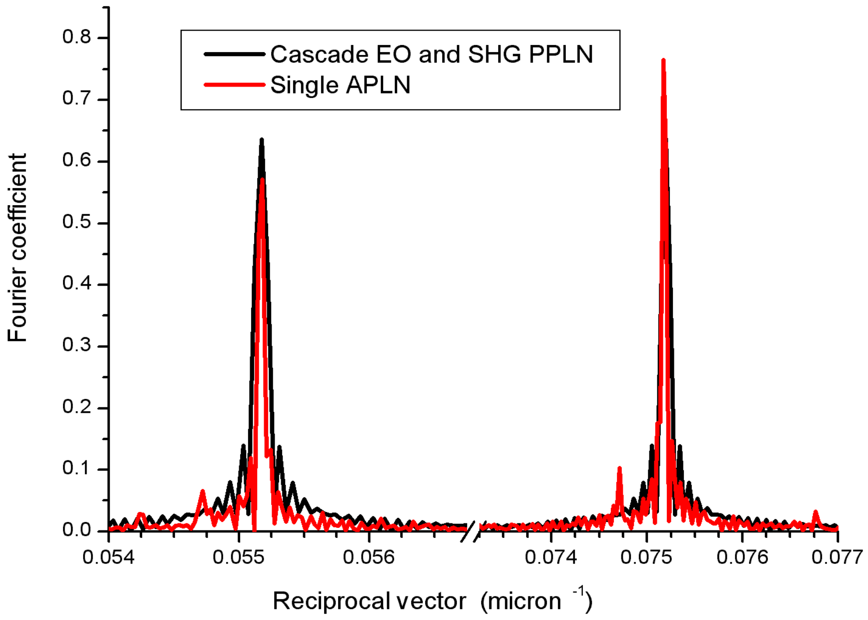


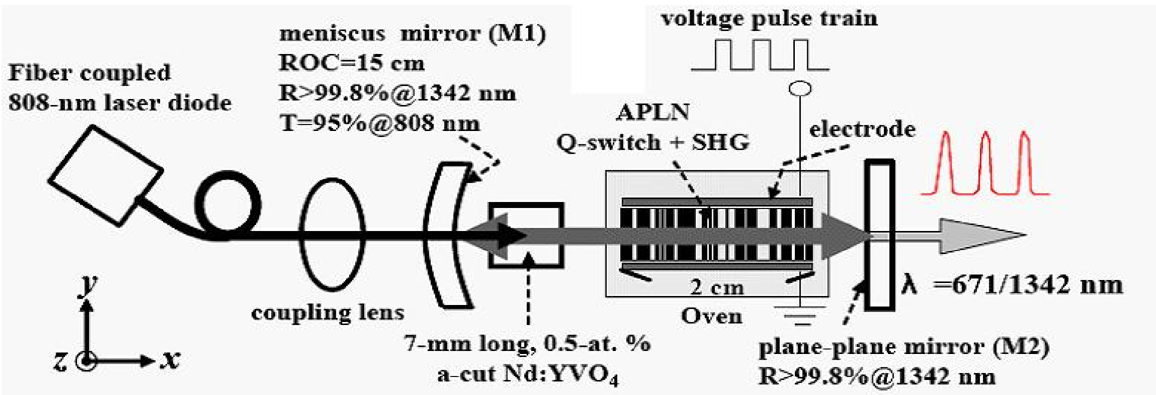
Fig. 5.3 The reciprocal vectors analyzed results are shown. There are reciprocal vectors of a 2-cm long bisection cascade PPLN which contents an EO PPLN and a QPM SHG PPLN (black curve), and that of a 2-cm long single APLN(red curve). it shows that there are only two dominant spatial frequencies which are mapping SHG and EO Q-switching respectively. The low frequency one ~0.055 μm^{-1} is mapping for EO QPM vector, and the high frequency one ~0.075 μm^{-1} is mapping for QPM SHG vector.

Fabrication

The fabrication process started from micro-lithography process. The pattern of APLN was predefined via spin-coated AZ4620 photoresist on +z surface of z-cut lithium niobate wafer (Crystal Tech. Inc.). Then the pattern of photoresist was transfered into lithium niobate domains through common electric poling technique. Poling electrodes were saturate lithium chloride solution on both z surfaces. Poling waveform was generated via a computer program controlled NI-DAQ interface adapter, and was amplified by a high voltage power supply (Trek Inc. Model 20/20C). Domain inversion responding electric current was monitored, and charge quantities were integrated to automatically terminate poling process for controlling the sideway expansion of inverted domains. Then HF solution etched the wafer to develop poled domain pattern for examination of quality.

Experiment

In the Fig. 5.4 it shows a schematic of a diode-pumped Q-switched 1342 nm wavelength laser resonator with a monolithic APLN frequency doubler and EO Q-switch in it. The pump laser is an 808 nm fiber-coupled diode laser with an emission polarization ratio of 2.5:1, and the polarization is coincident with a σ-polarized wave absorption band of a laser gain medium which is a 7-mm-long 0.5-at. %-doped a-cut $Nd:YVO_4$ crystal for end pumping. A coupling lens set is used to collimate and focus (with an effective focal length ~20 mm) the pump beam into the laser crystal near the front surface. The resonator is composed of two 15-cm radius-of-curvature (ROC) dielectric mirrors having > 99 % reflectance at 1342 nm and ~95 % transmittance at 808 nm. The cavity length of the constructed laser is as compact as 3.4 cm. The APLN chip is placed in the resonator on a temperature controlled oven next to $Nd:YVO_4$ crystal. The dimensions of APLN are 20 mm in length, 1 mm in width, and 0.5 mm in thickness. Both *x*-cut end facets are coated with AR coating at 808 nm, 671 nm, and 1342 nm. The *y*-cut facets are sputtered NiCr alloy film as electrodes for EO Q-switching. A power supply with voltage up to 500 V is cascaded a pulser and connected to electrodes on APLN to generate voltage pulse train.



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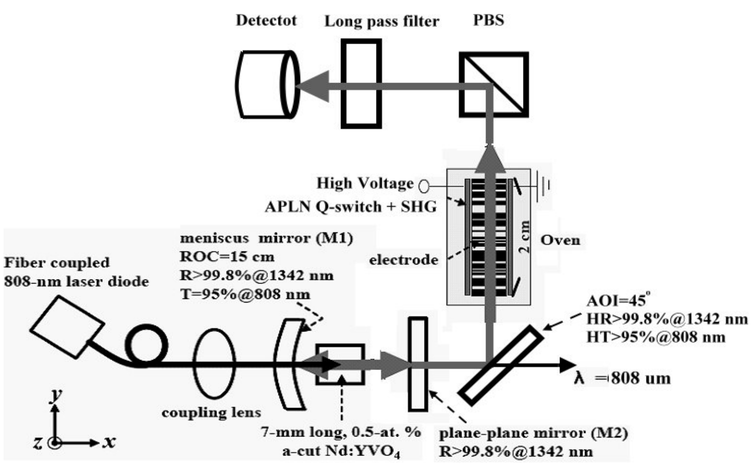
Before starting to operate the Q-switched laser, the APLN chip was placed out of the laser cavity to evaluate its PM conversion voltage and single path SHG performance. The corresponding scheme of setup is shown in Fig. 5.5. There were a long pass filter and an iris placed in front of detector to reject the 808 nm noise from pump laser. A polarization beam splitter (PBS) was used to extract the TE power. A 45º dichroic mirror with <5 % transmission at 808 nm and >99.8 % reflection at 1342 nm was placed behind laser cavity to separate pump power and the laser signal. The APLN half-wave voltage for phase-matching 1342 nm is shown in Fig. 5.6 which indicates that the measured voltage at ~37ºC is close to our design value, 390 V.



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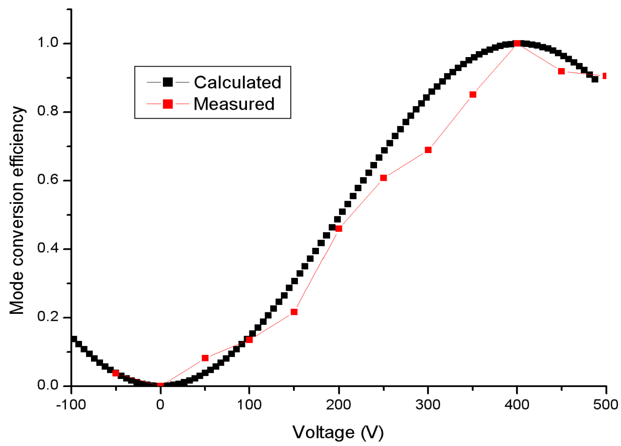


Fig. 5.6 The APLN half-wave voltage for phase-matching 1342 nm is shown which indicates that the measured voltage at ~37ºC is close to our design value, 390 V.

The normalized SHG efficiency versus temperature is shown in Fig. 5.7. It indicates that the performance is agree with the theoretical calculated values.

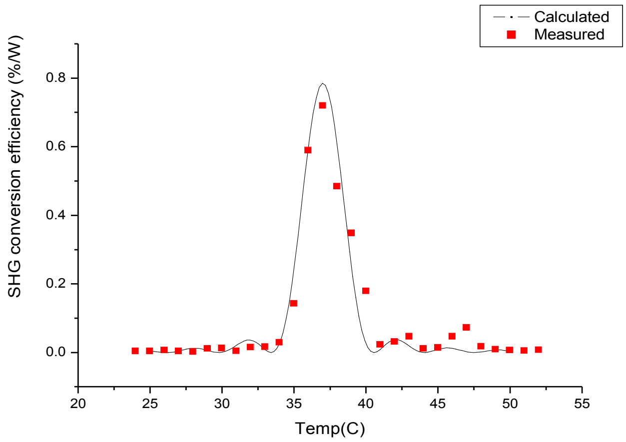


Fig. 5.7 The normalized SHG efficiency versus temperature is shown. It indicates that the performance is agree with the theoretical calculated values.

Then the APLN was moved back to the the laser cavity to operate the laser in Q-switching mode. Temperature was set at 37ºC. Applied voltage was under 195 V for 45º polarization turning. As shown in Fig. 5.8 for a driven pulse train voltage at ~190 V and repetition rate at 1 kHz a SHG-pulse width ~10 ns was measured, and there were ~25 % power depletion at 1342 nm.

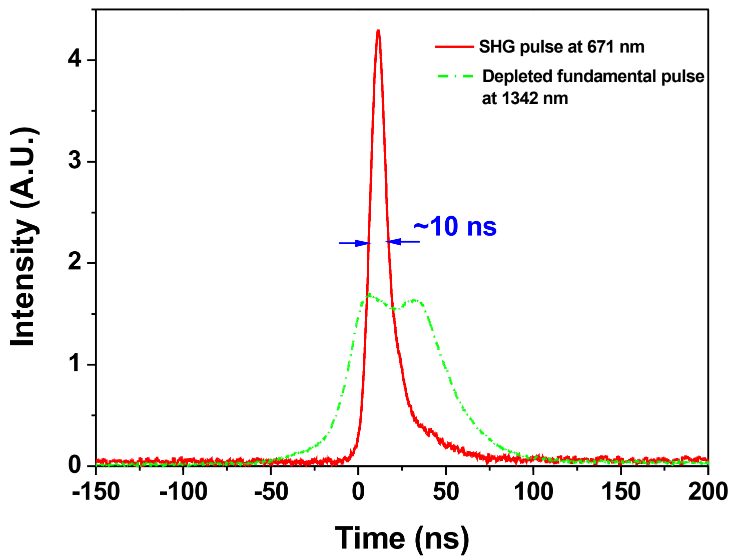


Fig. 5.8 For a driven pulse train voltage at ~190 V and repetition rate at 1 kHz a SHG-pulse width ~10 ns was measured, and there were ~25 % power depletion at 1342 nm.

Summary

In summary, an intracavity actively Q-switched Nd:YVO4 frequency-doubling laser have been successfully designed and experimental demonstrated with a single APLN that can be a low voltage active EO Q-switch and a QPM frequency doubler. It proved that the QPM wavelength conversion coupled-mode equations and EO QPM polarization rotation coupled-mode equations can be considered simultaneously with the aid of computational iterative method to generate a single AOS binary sequence. Reciprocal vector analysis shows that it is a efficient way to integrate multiple functions into a single chip for better performance. As driven pulse train repetition rate at 1 kHz a SHG-pulse width ~10 ns was measured, and there were ~25 % power depletion at 1342 nm.

In the future, this laser system can further incorporate into a Ti-diffused waveguide in rare earth ion doped lithium niobate single crystal substrate with high-reflectance coating on end surfaces to realize a TTL level pulse driven monolithic Q-switched frequency doubling laser as a light source for telecommunication and so forth.

Chapter 6 Gain-enhanced and spectral-narrowed optical parametric oscillator using PPLN electro-optic polarization-mode converters

Optical parametric oscillation has been one of the most promising techniques in producing wavelength tunable coherent radiations for various applications including spectroscopy, communication, and remote sensing. $LiNbO_3$ has been used to build optical parametric oscillator (OPO) since 1965 [48]. High nonlinear coefficient, wide transparency range covering the visible and extending into the mid-IR (from 0.35 μm to 5 μm), and capability for birefringent phase matching of useful interactions within its transparency window of $LiNbO_3$ that make it especially interesting for OPOs [19]. $LiNbO_3$ is one of the most famous optical ferroelectric materials. The maturity of the fabrication technique in quasi-phase-matching (QPM) materials such as the periodically poled lithium niobate (PPLN) crystal has led to a rapid advance of the OPO development. The first QPM OPO demonstration in bulk PPLN were published in 1995 [17]. The large effective nonlinear coefficient and free of walk-off problems in PPLN has facilitated it to construct an efficient OPO.

Most of the demonstrations in PPLN OPOs used the first-order QPM collinear interactions with all waves polarized extraordinarily to access the largest element of the nonlinear susceptibility tensor. For a first-order QPM PPLN device, the effective nonlinear coefficient is ~17 pm/V. The phase-mismatch of a specific QPM optical parametric process is given by

$$\Delta k_Q = \frac{n_p \omega_p - n_s \omega_p - n_i \omega_i}{c} - K_Q \tag{6-1}$$

where n_p, n_s, and n_i are the refractive indices of the pump, signal, and idler waves, respectively, c is the light speed in vacuum, and $K_Q \equiv \frac{2\pi}{\Lambda}$ is the grating vector of a domain period Λ. In the low-conversion limit, the single-pass parametric gain of a PPLN is around 20 times [17] higher than that of an unpoled birefringent $LiNbO_3$. Wavelength tunings in PPLN OPOs can be accomplished via adjusting the crystal temperature, angle, and grating period. Wide continuous wavelength tuning has also been demonstrated with the PPLN OPO technology by varying the PPLN temperature [18].

An additional distinctive advantage of PPLN is flexibility afforded by QPM technique allowable for engineering and tailoring a specific QPM domain structure. Thus high efficiency, wide wavelength tunable OPO is also possible to be constructed in a QPM crystal via the design of the grating structures implemented in it [20][49]. These tuning techniques have realized QPM OPOs very popular means to achieve broadly tunable coherent sources of radiation.

However, wavelength tuning in OPOs using either temperature tuning or QPM grating-vector tuning is mechanically slow, Though a rapid tuning mechanism in an ORO can be achieved via the wavelength tuning of the pump laser [50], the availability and quality of such a pump laser are problematic.

Tuning throughout the gain bandwidth of a PPLN OPO has been demonstrated by mechanical rotation of an etalon [20], electrical tunable Fabry–Perot interferometer [21], or mechanical rotation a blaze grating [22] within the OPO cavity. They also provide a spectral narrowing mechanism in the systems. However, these intracavity elements generally produce additional system loss as well as complexity and usually rely on a slow tuning mechanism using mechanics. Non-mechanical gain bandwidth tuning techniques based on the EO effect have been proposed using an asymmetric duty cycle PPLN [51], and a segmented PPLN crystal in OPOs [52]. These presented EO tuning methods offer the possibility of rapid wavelength tuning.

Currently the PPLN/APLN electro-optically active (EOA) PM converter (PMCs) have been successfully constructed and demonstrated as a laser Q-switch, as a multiple-wavelength filter, and as simultaneously a laser Q-switch and a SHG in in this dissertation work (see previous chapters). Since EO tuning has been proven as an effective method to tune the oscillation wavelength of an OPO, it is interesting to ask if it is possible to perform the wavelength tuning in a QPM OPO with

PPLN EOA PMCs? In this work, we further integrated PPLN EOA PMCs with a PPLN OPO in a monolithic substrate and demonstrated unique spectral manipulation of the OPO signal. We found the OPO output spectrum can be manipulated electro-optically by applying an EOA field and furthermore by temperature tuning. In particular, we have realized a gain-enhanced and spectral-narrowed OPO signal by this technique. The preliminary study results of such a unique system will be presented in this chapter. A reliable model for further design to achieve high efficiency, broad and fast tuning, and spectral manipulation in a compact OPO device is developing.

Device design and working principle

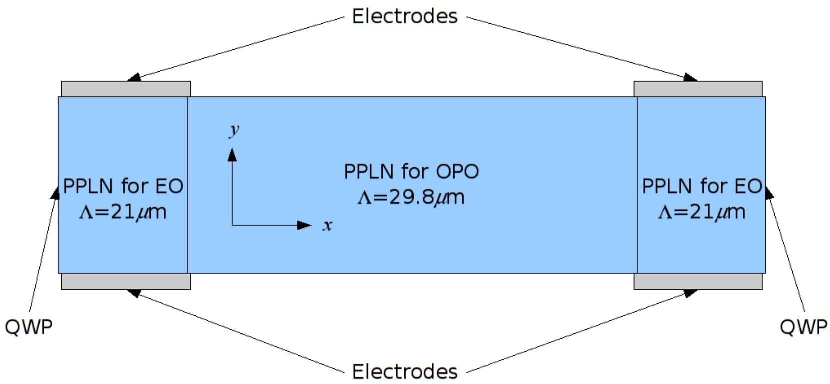
The essential idea of this work is to integrate the two functionalities of EOA filtering and optical parametric generation in a monolithic PPLN crystal to study the electro-optic control of the OPO output spectrum characteristics. Fig. 6.1 shows the schematic of the EOA OPG device design, where a PPLN optical parametric generator (OPG) is sandwiched between two PPLN EOA PMCs. The PPLN OPG is 3-cm long and has a grating period 29.8-μm phase-matched to the 1064-nm pumped 1532-nm signal and 3483-nm idler parametric generation process at 40˚C. The two identical PPLN EOA PMCs are each 1.25-cm long EO and have a grating period of 21-μm phase-matched to the codirectionally coupled Šolc-type Bragg condition for a 1532-nm wave at 40˚C. All these three PPLN devices are monolithically integrated in a *z*-cut 0.5-mm thick congruent-grown lithium niobate single crystal substrate, and the overall dimensions in *x*, *y*, and *z* directions of this monolithic device are ~55 mm, 1 mm, and 0.5 mm respectively. Each end in of the monolithic chip has a quarter-wave plate (QWP) section for ideally converting a polarization state from linear to circular. The QWP sections are required to function the two PPLN EOA PMCs as effective polarization rotators in a resonator [5].

Since the PPLN OPO produces a TM polarized signal wave, the parametric generated 1532-nm signal will become a circularly polarized wave after passing through the quarter wave plate when the EO PPLN is operated as a 45º polarization rotator. After being reflected at the cavity mirror, the wave again passes through the quarter wave plate and then the EO PPLN, which causes another 45º rotation of the 1532-nm signal wave. The reflected signal wave is thus a TE polarized wave and sees no gain in the PPLN OPO section. After experiencing the same action of rotation through the other end of the device, the signal wave recovers its polarization to the TM mode. In short, if the applied EOA voltage is $V_{\pi,PPLN}/2$ for the 1532-nm signal wavelength, the round-trip polarization change of the wave is ~360˚ and thus the oscillation can be built up. As a result, one will find that in this device the signal wave always sees parametric gain in one direction, and thus resembling a traveling-wave oscillator. A single-frequency optical parametric signal output can be expected. Due to the unidirectional gain, backward conversion will be suppressed and 100% parametric conversion efficiency is possible.

However, considering the existing but different bandwidths of the PPLN EOA PMC and the PPLN OPO, if the bandwidth of the PPLN EOA PMC is much larger than that of the PPLN OPO, then the aforementioned interesting mechanism can more or less work well. In current case, the 1.25-cm long PPLN EOA PMC has a bandwidth of ~1.03 nm [1] and the 3-cm long PPLN OPO has a calculated single-pass bandwidth of ~1-5 nm for a pulsed pump scheme (high-gain scheme). In such a situation, a more complex mechanism other than the traveling-wave OPO mechanism will be involved. We found experimentally it is the spectral filtering effect of the two PPLN EOA PMCs that domains the spectral performance of this unique OPO. For example, since the bandwidth of the PPLN EOA PMC is comparable or smaller than that of the PPLN OPO, for those non-phase-matched wavelengths in the bandwidth, their round-trip polarization changes after the PPLN EOA PMC will not be equal to 90˚, which results in an elliptically polarized light. Thus for a wavelength having a round-trip linear polarization change of equal to 180˚, the oscillation of this wavelength will not be built up. This leads that the PPLN OPO section plays also as a ultra-thick wave retard plate. It has been too complex to analyze the operation mechanism of such a system easily.

Therefore a theoretical model has to be developed to well describe such an interesting and unique system.

Modeling program has been constructed to simulate this novel OPO system with a monolithic PPLN comprising an OPG sandwiched by two EOA PMCs based on the QPM OPG coupling process in the PPLN and the EO QPM coupling process in the EO PPLN (refer to chap. 2) The output spectra of such a system were simulated to compared with the experimental data and will be presented in the following section. The pump depletion in the OPO system is not yet considered in this model for the first qualitative comparison. A model at considering the pump depletion scheme is developing under way for further quantitative comparison.



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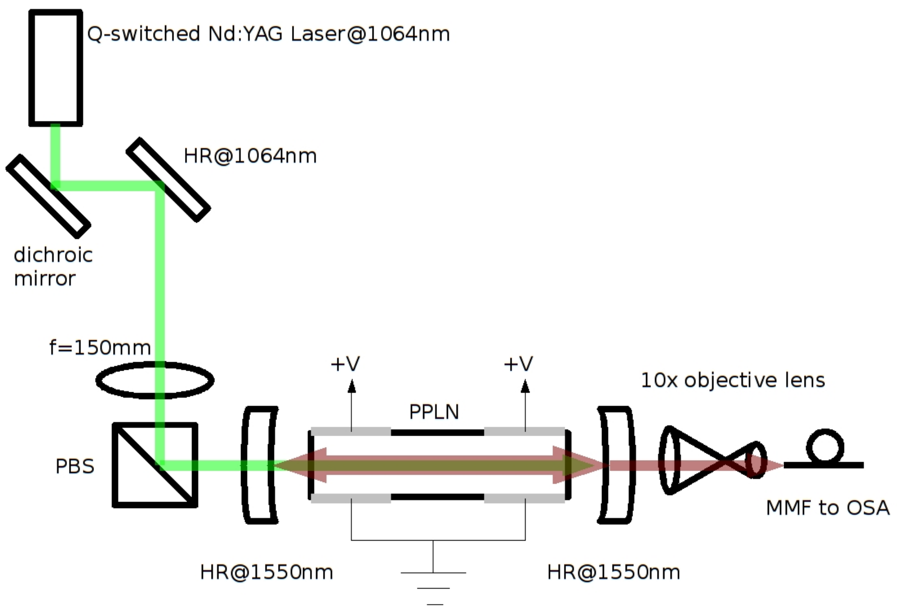
Experiments

The experimental setup is shown in Fig. 6.2. There was a lamp-pumped actively Q-switched Nd:YAG laser with 50 Hz repetition rate as a pumping source for the OPO device shown in Fig. 6.1. There were a steering HR 1064nm mirrors for beam alignments and a dichroic mirror as a filter to cut off unwanted wavelengths such as 532 nm and 354 nm radiated from the same laser. A polarization beam splitter was placed in the path to select the extraodinary light into SRO cavity. A beam coupling lens with 150 mm focus length was used to couple the 1064 nm pump laser into PPLN OPG section. Two identical concave laser mirrors having a 100-mm radius of curvature with ~99% reflectance at 1532 nm and high transmittance at the pump and idler wavelengths were placed apart to construct the resonator. The cavity length was about 10 cm. Both end faces of the PPLN EOA OPG crystal were polished and coated with anti-reflection (AR) dielectric layer for both 1064 nm and 1532 nm and applied with metal electric conductive paint as to the two *y* faces as the side electrodes. Then it was mounted in a temperature controlled oven with precision 0.1˚C in the resonator. The driving fields of the two PPLN EOA PMCs can be applied independently. The operating temperatures of the two EOA PMCs can also be separately detuned from the designated value 40ºC with our oven. The output SRO signal was collected by a 10x objective lens and a 1550 nm multi-mode fiber (MMF) connected to a a high spectral-resolution (~0.01 nm) optical spectra analyzer (OSA) (Advantech Inc. Model: Q8384). The swap mode of the OSA was synchronized via a trigger signal from the Nd:YAG pump laser.

Since a PPLN OPO provides parametric gain only for the phase-matched extraordinarily (TM) polarized wave and a PPLN EOA PMC can convert a phase-matched wave of TM-polarized mode to the TE-polarized mode, it is possible to select a certain spectral gain windows for the signal oscillation by the proper overlap of the PPLN OPO gain spectrum and the two PPLN EOA-PMC conversion spectra. This can be achieved by detuning away the phase-matching conditions of the two EOA PMCs by temperature and then by selecting the relative efficiency of the two conversion spectra by setting proper EOA fields for a desired spectral overlap. If a narrow spectral gain window can be created, gain enhancement and spectral narrowing of the OPO signal can be expected. Fig. 6.3 shows the measured (solid black line) and simulated (dashed black line) output spectra of the OPO signal for such a situation. The operation temperatures and applied EOA fields of the two EOA PMCs are 39.7/40.6ºC and 1200/1600 V/mm, respectively. In Fig. 6.3, the measured (solid red line) and simulated (dashed black line) OPO signal spectra without the application of the EOA fields are also plotted for comparisons. It clearly shows that the output signal is narrowed by a factor of 10 in spectral width and increased by a factor of 3.3 in peak intensity when the PPLN EOA PMCs are enabled. Fig. 6.4 further shows the measured and simulated results for producing 2 and 3 signal peaks in the OPO gain bandwidth using the present technique. Detailed system parameters for the results are given in figure legends.



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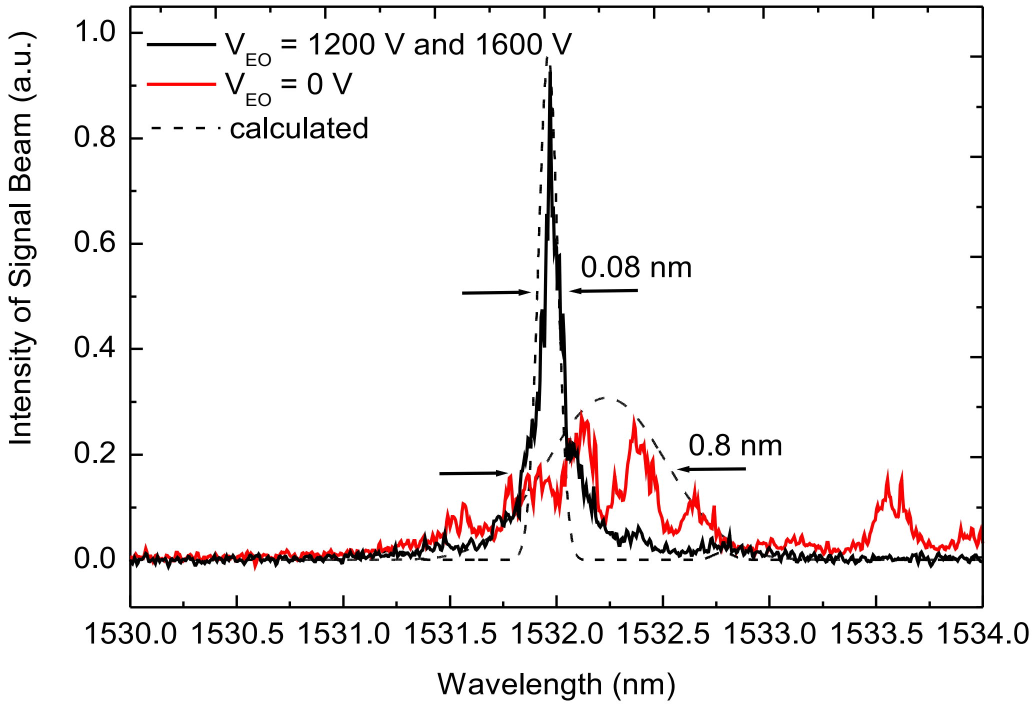

Fig. 6.3 Signal beam output spectra of the device with (solid black line) an without (solid red line) electric field applied to the EOA PMCs compared to simulated data (dashed lines).

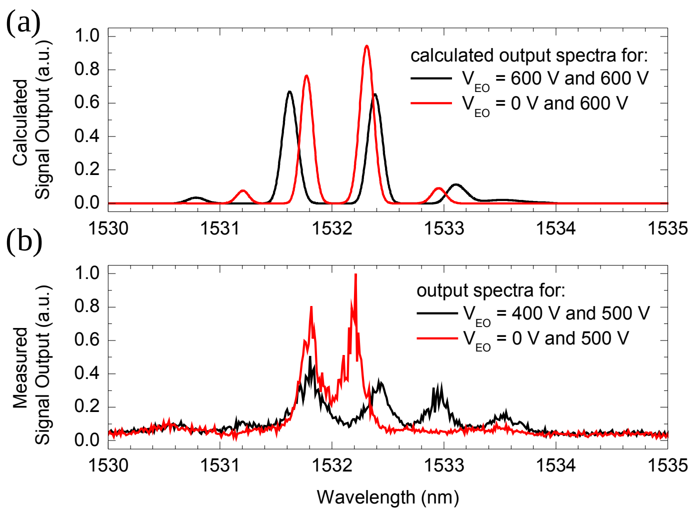

Fig. 6.4 (a) Simulated and (b) measured OPO signal output spectra, showing 2 and 3 signal peaks in the OPO gain bandwidth. Qualitatively good agreement between experiment and theory was found.

Conclusion

We have designed and constructed a novel OPO system with a monolithic PPLN comprising an OPG sandwiched by two EOA PMCs to achieve unique spectral manipulation of the output signal. OPO signal with a spectral width reduced by a factor of 10 and peak intensity increased by a factor of 3.3 was obtained by this system when the two EOA PMCs functioned at temperatures 39.7/40.6ºC and driving fields 1200/1600 V/mm, respectively. Such an EOA OPO technique has great potential for, e.g., communication and remote-sensing applications.

Chapter 7 Summary and Conclusion

In this dissertation work, the working principles, design methods, constructions, and experimental demonstrations of several novel photonics devices and lasers integrating high-efficiency QPM wavelength converters and unique electro-optically active (EOA) PM converters (PMCs) in monolithic periodically poled lithium niobate (PPLN) and aperiodically poled lithium niobate (APLN) crystals are presented. These devices and lasers are of potential for versatile applications including the optical communication and signal processing, remote sensing, biomedicine, displays, optical storage, ranging, etc.

In chapter 3, we have reported the design, fabrication, and demonstration of a laser-diode-pumped, actively internal-Q-switched laser system using a 17-mm long, 13.6-μm period monolithic Nd:MgO:PPLN crystal. We obtained 28-ns, 2.45-μJ Q-switched laser pulses, corresponding to a laser peak power of ~88 W, from this miniature 1.085-μm laser system with 2% output coupling when driven by a 5-kHz 260-V voltage pulse train and pumped at 0.61-W absorbed diode power. The characterized output performance of this laser system agreed reasonably well with the theoretical calculations. This demonstration reveals that this laser system has the potential to develop as an interesting internal-Q-switched self-frequency-conversion laser if an additional QPM wavelength-conversion section can be monolithically integrated into the current Nd:MgO:PPLN crystal.

In chapter 4, we have successfully demonstrated active multi-wavelength optical filters by using EO APLN crystals. We derived a coupled-mode theory to model the device behavior and developed a computer code based on the simulated-annealing algorithm to optimize the design of such EO APLN filters. In experiment, we simultaneously transmitted 8 ITU-standard wavelengths through a 5-cm long EO APLN filter with a transmittance of >90% and a bandwidth of ~0.45 nm for each wavelength channel. The transmission spectrum of the EO APLN filter can be conveniently tuned by temperature at a rate of ~0.65 nm/ºC in the telecom C band. To operate at the TTL-level voltage, the EO APLN filter presented in this work could be implemented in, for example, a Ti-diffused $LiNbO_3$ waveguide.

In chapter 5, we have developed an electro-optically Q-switched intracavity SHG laser by using a single-grating-structure APLN device. A proof-of-principle demonstration of a pump Q-switched intracavity SHG $Nd:YVO_4$ laser by using the APLN crystal as simultaneous a laser Q-switch and a SHG. was performed. Both theoretical and experimental results show that the single-pass SHG conversion efficiency of this APLN Q-switch-SHG device can be >1.5 times higher than that of the cascade PPLN Q-switch and SHG device under the same crystal length and applied electric field. When the system was pumped at a diode power of 2 W and the APLN Q-switch was driven by a 120-V pulse train at 1 kHz, we observed a highly pump-depleted laser pulse at 1342 nm and a sharp SHG pulse at 671 nm with a pulse width of ~10 ns. A ~25% energy depletion of 1342-nm fundamental wave was estimated, indicating the high efficiency of such a device. An internal-Q-switched self-frequency doubler Nd:APLN laser will be pursued next.

In chapter 6, we further integrated PPLN EOA PMCs with a PPLN OPO in a monolithic substrate to achieve unique spectral manipulation of the OPO signal. A novel EO OPO system, pumped by a pulsed 1064 laser, comprising a 3-cm long, 29.8-μm PPLN OPG sandwiched by two 1.25-cm long, 21 μm PPLN EOA PMCs was designed and constructed. OPO signal at 1532-nm with a spectral width reduced by a factor of 10 and peak intensity increased by a factor of 3.3 was obtained by this system when the two EOA PMCs functioned at temperatures 39.7/40.6ºC and driving fields 1200/1600 V/mm, respectively. Using present technique to produce 2 and 3 signal peaks in the OPO gain bandwidth was also demonstrated. Such an EOA OPO technique has great potential for, e.g., communication and remote-sensing applications.

I believe the nonlinear integrated phonics devices and lasers initiated and developed in this work will lead to more extensive studies on relevant research topics and invent more compact, efficient, and advanced phonics systems for versatile applications and contribute to the science and technology development on integrated optics, lasers and nonlinear optics.

Chapter 8 Outlook

The central idea and objective of this study is at the development of attractive monolithic multi-function devices via the integration of the unique electro-optic (EO) effects of a quasi-phase-matching (QPM) material with its capability of performing efficient nonlinear frequency conversion. However, besides the QPM and EO technologies, the proper incorporation of the impurity doping, distributed-Bragg-grating, and even acousto-optic (AO) techniques in nonlinear materials could lead to the construction of powerful all-optical devices. These devices are capable of integrating multiple functions in a monolithic crystal, including laser generation, wavelength conversion, amplification, modulation, switching, multiplexing, filtering, and so on.

Extended from the ideas for those nonlinear integrated photonics and laser devices demonstrated in this study, several new devices can be derived:

A. New schemes derived from the Nd:MgO:PPLN laser (chapter 3) and the pump Q-switched intracavity SHG using a single APLN in Nd:YVO_4 laser (chapter 5)

1. A monolithic electro-optically active internal-Q-switched, self-wavelength-conversion Nd:MgO:PPLN system can be constructed if an additional QPM wavelength converter such as SHG/SFG/OPG can be monolithically integrated into the Nd:MgO:PPLN crystal.

2. A pump Q-switched intracavity OPO using a single APLN in Nd:YVO_4 laser can be constructed.

3. A monolithic electro-optically active internal-Q-switched, self-wavelength-conversion Nd:MgO:APLN system can be constructed.

B. New schemes derived from the active multi-wavelength optical filters in EO APLN (chapter 4)

1. A monolithic device integrating a multi-channel EOA filter with a multi-channel SHG/DFG in a single-structure APLN can be constructed. Such a device will be of interest in data-communication

and encryption-communication.

C. New schemes derived from the EOA PPLN OPO system (chapter 6)

1. A two-mirror traveling-wave SRO can be constructed provided a spectral narrower element can be used, such as a volume Bragg grating (VBG) as its cavity mirror.

2. A single-frequency SRO in a linear cavity comprising a PPLN OPG sandwiched by two *contradirectionally* coupled EOA PPLN reflectors can be constructed.

All of the demonstrated and proposed devices in this dissertation are also favorable to be implement in waveguide devices. With the waveguide configuration the voltage demanded for EO operation can be down to the TTL level and therefore can be driven in a much higher switching rate. Furthermore, it would be much easier to integrate/couple with fiber based systems. To achieve this goal a stable waveguide fabrication process in PPLN/APLN has to be established.

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